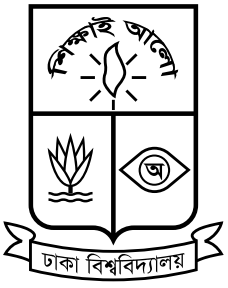
UNIVERSITY OF DHAKA

Department of Computer Science and Engineering



AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

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Declaration Form

This progress report presents the research foundation, current implementation, and development-stage experimental evidence for the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. The work is conducted by the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets, software libraries, and findings originating from prior work are acknowledged and cited. The numerical forecasting results are restricted to the supplied daily shop–category panel and the development protocol described in this report. They do not represent a final untouched-test evaluation or a claim of effectiveness across all SME sectors. This report is current to 8 November 2026; the project remains in progress until February 2027.

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Abstract

This thesis develops B-SMART, an AI-powered business analytics and decision-support framework for performance evaluation, demand prediction, customer insight and feasible recommendations in Bangladeshi SMEs. The supplied dataset manifest identifies the business package as real SME sales and real people. The RQ1 daily panel contains 561,152 observations across 32 shops, 16 categories and 512 shop–category series from 2023 to 2025, with 1,580,361 observed units and 37.60% zero-sales observations. The forecasting target is observed sales, not unobserved latent demand.

A leakage-controlled LightGBM Tweedie experiment compared historical series features with additional shop, category and network context features. Across nine monthly seven-day development origins, pooled WAPE decreased from 33.47% to 32.66%; the context model won six of nine origins. Its aggregate bias increased in magnitude from -2.02% to -6.05%, illustrating the trade-off between absolute error and under-forecasting. The context model also improved pooled WAPE in the three-origin 1-, 14- and 28-day development comparisons. Prequential interval coverage and bottom-up hierarchical aggregation were evaluated as additional diagnostics. These results are development evidence, not a final untouched-test claim. SKU-level latent demand, inventory-policy benefit and customer-action effectiveness require separate evidence and evaluation.

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Chapter 1

Introduction

1.1 Context and Motivation

SMEs generate valuable evidence through sales, purchases, payments, inventory movements and customer transactions, but these records are often distributed across ledgers, spreadsheets and separate applications. Owners may see that sales have changed without knowing which products, customers or operating conditions explain the change, or what response is feasible.

The challenge extends beyond displaying historical KPIs. Retail decisions require demand estimates while accounting for cash, stock, supplier lead time, pack size, storage and expiry. Customer inactivity, incomplete cost records and promotion effects create further analytical risks. This research connects operational events to valid predictions and explainable, constraint-aware decisions [4, 8, 17].

1.2 Bangladesh SME Business Context

The intended setting is Bangladeshi SMEs in retail, wholesale, distribution and related sectors. A shared operational core covers sales, purchases, expenses, payments, customers and performance indicators; inventory, batch, expiry and capacity modules apply where relevant. A common framework does not imply that one model suits every business.

Bangladesh studies identify management support, perceived benefit, cost, complexity and organisational readiness as important influences on SME technology adoption [11, 22]. These findings motivate low-friction data entry, local payment compatibility and owner-readable evidence. Results remain bounded to the businesses and periods represented in the supplied data.

1.3 Decision Environment and Stakeholder Requirements

The owner or manager makes the principal decisions, while operational, finance and supply staff create or validate supporting records. The system must preserve traceability from each result to its source events, transformation rules and historical cutoff. It must distinguish incomplete evidence from a valid zero and allow owners to inspect, modify or reject recommendations.

Table 1.1: Core stakeholder requirements.

Stakeholder	Responsibility	System requirement
Owner or manager	Reorder, allocate cash and approve customer actions	Feasible options, uncertainty, supporting evidence and owner control
Operational users	Maintain product, sales and stock records	Stable identifiers, validation and correction history
Finance personnel	Reconcile payments, expenses and costs	Traceable transaction links and missing-cost coverage
Supplier contacts	Confirm lead time, pack size and incoming quantity	Dated supply constraints, not assumed universal values
Research and system team	Build and evaluate the artifact	Versioned data, code, cutoffs and reproducible results

The design is assessed on decision relevance, traceability, proportionality and contestability.

1.4 Problem Statement

The central problem is the weak connection between SME operational records, valid analytical estimates and feasible business actions. Four gaps follow:

G1. **Data foundation:** records need consistent identifiers, timestamps, provenance and business boundaries.

G2. **Prediction validity:** features, labels and transformations must respect decision-time availability and chronological evaluation [15].

G3. **Prediction to action:** forecasts and risk scores must be linked to cost, stock, cash, lead-time and consent constraints [1].

G4. **Adoption and accountability:** owners need understandable, traceable evidence and authority to reject or modify recommendations. Feature attribution supports inspection but does not establish causality [16, 18].

The thesis evaluates an integrated SME decision-support framework. It does not assume that machine learning must outperform every baseline; a simpler method is valid when more reliable or economical for the task.

1.5 Analytical Logic and Claim Boundaries

The process separates an observed business event, an estimate based on information available at that time, an eligible action and a later outcome. Missing records weaken the estimate; leakage invalidates it; ignored constraints make an action infeasible; and absent follow-up prevents a credible impact claim.

The thesis distinguishes descriptive KPIs, predictive estimates, prescriptive recommendations and realized business impact. Recorded sales may be censored demand when stock is unavailable. Future inactivity means no observed purchase within a defined window, not permanent customer loss. Model explanations describe model behaviour, not causal effects [16, 18]. Claims are limited to the businesses, periods and tasks actually evaluated.

1.6 Research Aim and Objectives

1.6.1 General Objective

The overall goal is to create and test an AI-based business solution. The analytics framework is designed as a solution that allows Bangladeshi SME businesses to track their business performance, extract reliable and explainable insights, and prioritise feasible strategies while maintaining owner control.

1.6.2 Specific Objectives

The study will:

O1. specify an auditable canonical data model for sales, with a focus on the business and its activities, including products, stock, batches, expiry, purchases, returns, customers, suppliers, payments, and expenses;

O2. build reproducible preprocessing and feature pipelines for performance analysis, SKU demand, customer inactivity, and segmentation;

O3. compare transparent baselines, a pooled multi-business model, and shop-adapted models using chronological and held-out-business validation, with both predictive and decision-oriented measures;

O4. measure uncertainty and provide explanations at both global and case levels, while recognising that association does not imply causation;

O5. formulate a mechanism that retains constraint awareness in reorder, retention, and resource-allocation decisions; and

O6. assess technical correctness, robustness, scalability, usability, and potential business value at distinct levels of evidence, using data collected with consent from the Bangladeshi SME sector.

1.7 Initial Solution Concept

The proposed solution is B-SMART, a multi-sector SME decision-support framework with a shared operational core and configurable analytical tasks. It validates business events, creates time-frozen analytical views, compares baselines with eligible models, and presents estimates with their cutoff, uncertainty and supporting evidence.

Its three functions follow the approved project title: *performance evaluation* reconciles sales, stock and financial indicators; *insight extraction* identifies demand patterns, customer inactivity and operational anomalies; and *strategy optimisation* ranks feasible purchasing or customer actions under business constraints. The owner retains final authority, and outcomes are recorded for audit and monitoring.

1.8 Research Questions

The thesis is based on four research questions:

RQ1.
Do features such as calendar effects, price, promotion, and market context enhance out-of-time branch–SKU demand forecasts over seasonal and statistical baselines in the target business setting?

RQ2.
How accurately and reliably can a historical customer snapshot predict whether there will be no repeat purchase within a predetermined future time period?

RQ3.
Under the same demand and operating constraints, can a forecast-driven reorder policy reduce stock-outs and holding costs compared with the current policy of the participating business?

RQ4.
Do short Bangla action descriptions enhance owner task-friendliness, confidence, and perceived usability without causing information overload or increasing decision time?

1.9 Scope and Boundaries

Empirical claims are restricted to the supplied Bangladesh business dataset and tasks for which its fields are suitable. The current forecasting experiment is at shop–category–day level and predicts observed sales; it does not establish SKU-level latent demand. Customer-risk analysis requires stable pseudonymous identifiers and complete follow-up. Inventory-policy claims require reconciled stock, cost and supplier constraints. Cross-sector generalisation and realized business impact require separate evaluation.

1.10 Expected Contributions and Significance

The evaluated integration of established research methods, constrained within the context and relevance of small SME businesses, is expected to be a research contribution. Specifically, the work is expected to include (i) a traceable business operational-to-analytical data design; (ii) reusable historical-snapshot protocols designed to avoid common forms of data leakage; (iii) a Bangladesh-aware comparison of forecasting and customer-inactivity approaches; and (iv) a clear link between prediction and constrained, actionable recommendations.

The work also provides evidence of the ability of owners to interpret and apply the information presented to them. When making a recommendation, the system presents supporting evidence to help the owner understand the "how and why" of the recommendation, while using familiar rules and maintaining the owner's control over the action. It is not intended to exclude the owner from the decision or to label a numerical output as profitable without considering the business context. Before measuring the impact on the business, the system must first establish that its outputs are accurate, interpretable, and actionable.

1.11 Anticipated Challenges

The main challenges are incomplete fields, stock-ledger anomalies, uneven demand, business heterogeneity, changing conditions and limited independent businesses. These are addressed through eligibility rules, baseline comparisons, temporal validation, uncertainty reporting and proportionate claims. Insufficient evidence is reported as not evaluable, not replaced with assumptions.

1.12 Organization of the Report

Chapter 2 builds up the technical and contextual background, reviews the relevant literature, and identifies the research gap. Chapter 3 introduces the research design, data protocol, formal tasks, proposed framework, and evaluation plan. The Implementation and

Experimental Results chapters are kept as placeholders in this preliminary report. The Conclusions chapter brings together only what can be concluded from the current background study and proposed methodology. Final conclusions based on empirical outcomes will be drawn from the evidence obtained during the implementation and experimental evaluation.

Chapter 2

Background Study and Related Work

2.1 Background Study

2.1.1 Business Intelligence and the Decision Continuum

Business intelligence (BI) is a blend of data, infrastructure, analytical techniques, best practices, experience, and organisational practice used to enhance decision-making. Chen, Chiang, and Storey trace its evolution from enterprise reporting to web, mobile, and sensor-rich analytics [4]. This thesis follows a three-step approach: *descriptive analytics* determines what happened and where; *predictive analytics* predicts what is likely to happen next; and *prescriptive analytics* compares feasible responses.

The stages are interconnected. An action recommendation is only as believable as the prediction that supports it, and a prediction is only as believable as the operational data from which it was obtained. There are also different integrity requirements for operational and analytical workloads. While a sale requires consistent updates to stock and payment records, an analytical model requires a historical feature view containing values that would not change once the evaluation point has been reached.

The proposed framework therefore separates the transaction record from versioned analytical snapshots. It maintains the connection between them through source identifiers, timestamps, and transformation metadata.

2.1.2 SME Digitalization in Bangladesh

The decision to adopt technology is not only a matter of the accuracy of technology models. A mixed-method study indicates that the important factors determining ICT adoption among SMEs in rural Bangladesh are awareness of the benefits, government support,

top-management support, and financial support [11]. A later survey of 535 SME managers combines the technology–organization–environment model with diffusion-of-innovation concepts. The authors [22] recognize that factors such as relative advantage, complexity, perceived cost, management support, competitive pressure, and government support are important considerations. These studies suggest that organizational readiness and costs, as well as technological fit and workflow functionality, warrant attention when considering technology adoption. However, the two studies do not evaluate a retail prediction system.

This operating context also includes mobile financial services. Through in-depth interviews, accessibility, spatial and temporal mobility, and access to information were found to be important issues in the daily use of MFS in Bangladesh. Self-control and the affordances related to self-control are also important considerations [9]. Therefore, the proposed data model includes the payment mode in the settlement status field, along with other common business fields. The study should not be construed as providing evidence of better business performance for a named provider or of better performance of one local provider compared with another.

2.1.3 Shared SME Core and Sector-Specific Requirements

The research scope covers SME businesses across sectors. Shared entities support comparable performance indicators, while sector-specific extensions capture differences in products, service capacity, sales cycles, procurement and operating constraints. Inventory-based firms may require SKU, stock, pack, batch and shelf-life fields; service firms may require service type, bookings, staff availability and delivery capacity. Fields that do not apply are marked not applicable rather than populated with invented values.

The artifact supports business performance evaluation, insight extraction and feasible strategy recommendations. Owners retain authority over actions. Sector-specific requirements and owner constraints are assessed before an analytical task is enabled; unsupported tasks return an explicit evidence gap.

2.1.4 Performance Evaluation and Data Foundations

Well-defined measures are the starting point for performance evaluation. Revenue, gross margin, average order value, stock turnover, return rate, and customer repeat rate each need to include a defined time window, an aggregation method, and appropriate treatment of events such as rescheduling, refunds, and lost fees. If any of these elements are not incorporated into a dashboard value, the resulting measure may be misleading or ignored. Definitions may also differ from the accounting records or may refer to different time periods. The canonical layer therefore maintains invoice lines and stock movements while creating analytical measures through documented views.

Data provenance concerns where a record came from, when it was received, and how it was transformed. Primary business data should be accompanied by provenance information,

together with appropriate customer anonymisation and organisational anonymisation. For public data, the context and reasonable implications of the applicable licence should also be documented.

2.1.5 Data Quality, Provenance and Temporal Semantics

Analytical validity depends on the meaning and timing of each field. Event time, recording time, correction time and decision cutoff are not interchangeable. Historical features must use only information available by the cutoff; later corrections or outcomes must not leak into earlier snapshots [15]. Missing records are not automatically zero, and unknown stock exposure remains unknown.

A reproducible manifest records source, owner, collection period, schema, checksum, row counts, exclusions and transformation rules. Reconciliation checks keys, duplicates, quantities and links among orders, sales lines, payments and inventory movements. These checks establish internal consistency; source documentation establishes declared origin.

2.1.6 Customer Analytics: RFM, CLV and Segmentation

A simple summary of *recency*, *frequency*, and *monetary value* is provided by RFM. It summarises a customer's purchase or transaction history using features that are easy to calculate and maintain an unambiguous behavioural interpretation. Fader, Hardie, and Lee link RFM with Customer Lifetime Value (CLV) through the Pareto/NBD and gamma–gamma components, demonstrating how the value of a particular customer history can be estimated [7]. CLV can be used as an outcome or planning measure, but it is not necessarily suitable as a direct prediction target or operational goal.

A field that defines a segment should not be admitted as an independent predictor of that same segment, as this would introduce information leakage. The question for unsupervised segmentation is whether coherent behavioural groups can be identified without an existing label. K-Means aims to:

$$\underset{\{C_k,\mu_k\}_{k=1}^{K}}{\text{minimize}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \tag{2.1}$$

The procedure returns clusters even when the data form a continuum. Therefore, both the number and the interpretation of the clusters need to be validated. The silhouette coefficient measures the similarity of observations within a cluster and their separation from the closest other cluster [20]. The stability of the clustering parameters across resampling and the business usefulness of the resulting clusters should be treated as separate tests. An attractive label such as "loyal" does not, by itself, imply statistical stability or actionability of a cluster.

2.1.7 Retail Demand Forecasting

A demand forecast estimates the number of units needed for a specific product, location, and future horizon. Retail demand series can be influenced by price, promotion, stock-outs, festivals, assortment changes, and aggregation across products or branches. Fildes, Ma, and Kolassa emphasize that retail forecasting practice cannot be simplified into a single algorithm because of data hierarchy, operational requirements, and the amount of data involved. The forecasting process and the way forecasts are used also matter, alongside statistical accuracy [8].

Intermittent SKU demand contains numerous zero-demand periods between non-zero observations. Croston's method smooths the non-zero demand size and the interval between demands separately. Syntetos and Boylan examine the accuracy and bias of intermittent-demand estimates [24]. This work supports the use of an intermittent-demand baseline rather than requiring a uniform and constant baseline. For dense time series, an appropriate dense-series model can be used at the SKU level. Forecasts also need to be reconciled across different levels of aggregation, including product, category, branch, and overall business totals. Optimal-combination reconciliation provides a principled means of generating coherent hierarchical forecasts [12].

When modelling, the model should be only as complex as the available evidence supports. ARIMA-family models represent serial dependence, including dependence after appropriate transformations; tree ensembles can learn nonlinear effects from lagged, calendar, and price variables; and sequence models can use ordered historical observations. Random Forest and XGBoost provide suitable tabular alternatives [2, 5]. The M5 retail competition illustrates the importance of large-scale, hierarchical, and uncertainty-aware comparison [17]. A seasonal-naive forecast is therefore a necessary benchmark rather than a ceremonial baseline.

Although inventory usefulness is related to forecast accuracy, the two are not the same. Hyndman and Koehler discuss the limitations of common percentage-error measures and suggest alternative approaches, including MASE as a scale-free measure for comparison [13]. A study by Syntetos and colleagues associates the choice of forecasting methods with stock-control requirements, whereas forecasting selection based solely on statistical accuracy may not reflect the needs of inventory management [23]. The methods are therefore evaluated using point-forecast error, uncertainty, and simulated stock-out and holding costs rather than selecting a technique based on a single easily computed metric.

2.1.8 Forecastability, Demand Regimes and Decision Loss

Retail series differ in volume, variability, intermittency, trend and seasonality. ADI and squared coefficient of variation describe regimes, but labels must use only history available at the forecast origin and should diagnose rather than automatically route models. Croston-family methods and TSB are relevant intermittent-demand comparators [6, 24, 27].

Forecast accuracy and inventory performance are distinct. Lower WAPE may not reduce shortages or holding cost when bias, lead time, pack multiples, cash and stock constraints are ignored. Models should be compared on common origins and targets, then evaluated through a separate policy replay [23].

2.1.9 Customer Inactivity, Class Imbalance and Calibration

Churn can be a documented cancellation for a subscription-based service. For non-contractual retail, however, churn has to be specified at a cutoff time t as no repeat purchase within a given time interval $[t, t + H]$. All features must be known at time t, and the training and validation processes must not be contaminated by information from the future label window. This definition translates a business term into a repeatable prediction task.

In business-to-business settings, the positive class can be scarce or vary across datasets. Saito and Rehmsmeier provide a rationale for the superiority of precision–recall analysis over ROC analysis for imbalanced binary classification tasks [21]. Model comparison should therefore report class prevalence and the quality of the predictions before considering action costs and benefits.

The prescriptive layer includes an additional requirement. When using a model output in a decision calculation, ranking alone is not sufficient; the output needs to be converted into an interpretable probability or score. A probability of 0.7 should, under appropriate calibration, correspond approximately to a 70% frequency of the event among comparable cases. Niculescu-Mizil and Caruana examine the calibration of probability estimates from supervised learners and post-hoc calibration methods [19]. The proposed framework therefore reports Brier score in addition to PR-AUC, ROC-AUC, and top-fraction lift, while also making calibration information available.

2.1.10 From Customer Score to Responsible Retention Decision

A customer-risk model does not grant permission for an action; instead, it provides a ranking or probability. For a given contact capacity, ranking quality determines whether the selected fraction contains more future inactive customers than an untargeted policy. When an expected value is calculated, the probability needs to be calibrated because the benefits and costs are subject to uncertainty and are weighted by the estimated probability. Therefore, the operational threshold should be determined using validation data and the pre-declared assumptions. The final test set should not be used to determine contact capacity or utility assumptions; rather, it should be used to confirm the choices made during the development process.

Additional ambiguity is created by the customer record. The frequency and amount of purchases may be linked to opportunities to visit, household purchasing patterns, or shared identifiers. Neither return behaviour nor discount-seeking behaviour alone is

sufficient to indicate loyalty or future value. Fader, Hardie, and Lee demonstrate why combined transaction history can provide more information than a single RFM label [7]. Segmentation can support interpretation, but a persuasive "segment name" is only a description of a behavioural pattern. It does not establish that the pattern is statistically stable. Therefore, clustering should be accompanied by stability checks and owner review before the resulting segment output is used for decision-making [20].

An independent feasibility gate is also required for responsible use. A score indicating a high probability of inactivity does not, by itself, justify customer contact if consent, contact policy, or an appropriate offer is absent. The score should also never be interpreted as a causal attribution, such as "long recency increased risk," or restated as a causal claim that a particular message will retain the customer. The thesis therefore treats the prediction of inactivity, the reliability of the estimated probability, contact eligibility, and subsequent customer response as separate links in the decision process. This structure allows the model to be scientifically informative even when the available evidence is insufficient to support a claim about profit or the causal effect of a retention action.

2.1.11 Leakage, Explainability and Change Over Time

Data leakage occurs when training is exposed to information that would not legitimately exist at prediction time. It can enter through target-derived features, customers who appear on both sides of a data split, post-customer-action information, post-outcome variables, or transformations that are fitted before the data are split [15]. Leakage control is therefore a characteristic of the data pipeline, not a model option. Before training, the system should define the prediction time, feature-availability rules, and data-splitting policy.

SHAP provides additive feature contributions relative to a reference value and supports both local and global inspection of model predictions [16]. Such an attribution describes how the fitted model uses encoded features; it does not establish a causal relationship. In social-science research, as synthesized by Miller, the focus is often on useful explanations that are selective and contrastive [18]. Accordingly, a recommendation card should respond to the question, "Why this action now?" by presenting a small number of salient, decision-relevant factors. It should avoid causal language while providing sufficient supporting evidence for the owner to assess the recommendation.

Business conditions also evolve over time. Retail monitoring therefore needs to distinguish between data-quality failure, input-distribution shift, and deterioration in outcome performance. A model should not be considered valid simply because it remains on a retraining schedule. Evidence that a model remains valid should come from continued evaluation under a declared validation protocol. Any retrained model should be evaluated using the same validation principles that were established during development, with changes explicitly documented.

Once a prediction is made, the next step is to take an action subject to business constraints. Prescriptive analytics incorporates predictions into an explicit decision problem. Bertsimas and Kallus develop prescriptions based on data with auxiliary information, where observations are used to inform conditional stochastic optimization [1]. The present thesis follows the underlying principle of evaluating actions according to expected benefit, cost, and constraints. It does not claim to reproduce the asymptotic method or theoretical approach of the cited work. Instead, reorder and retention decisions are treated as separate action families because their outcomes, costs, permissions, and uncertainties differ.

The proposed system is both an information-system artifact and an analytical study. Design-science research develops and evaluates artifacts in relation to a relevant problem and an existing knowledge base [10]. This perspective contributes to an evaluation plan that considers predictive validity, software behaviour, usability, and the consequences of decisions made by the owner, rather than treating a single model metric as the definitive criterion of success.

2.1.12 Cross-Stream Evidence Synthesis

The literature supports five connected requirements: reliable operational data, time-valid prediction, appropriate baselines, constraint-aware action and accountable owner use. Bangladesh adoption studies inform local design; forecasting and customer-analytics studies inform model evaluation; leakage, calibration and explanation research inform validity; and prescriptive analytics links estimates to decisions. The contribution is their integration and evaluation within the available SME evidence, not a newly invented algorithm.

2.2 Literature-Review Procedure

This is a focused background review, not an exhaustive systematic review. Sources were selected for relevance to Bangladesh SME digitalisation, business analytics, customer behaviour, retail forecasting, inventory, leakage, calibration, explainability and prescriptive decisions. Peer-reviewed studies, established methods and identifiable dataset repositories were prioritised. These sources provide contextual and methodological foundations; they are not substitutes for the empirical business records analysed here.

2.3 Research Gap and Thesis Positioning

Six connected gaps motivate the study:

G1 – Contextual evidence.
Bangladesh SME adoption research explains readiness and barriers, while large retail studies provide forecasting evidence; neither alone establishes transaction-level effectiveness in the target operating context.

G2 – Operational integration.
The thesis preserves the path from source event and correction history to frozen snapshot, prediction, action and outcome.

G3 – Joint methodological validity.
Leakage control, intermittent demand, hierarchy, forecast metrics and calibration must be applied together.

G4 – Prediction to decision.
Forecasts and customer scores must be translated into feasible actions under cash, stock, lead-time, cost and consent constraints.

G5 – Owner-centred explanation.
Feature attribution alone does not show that an owner understands or appropriately uses a recommendation.

G6 – Lifecycle accountability.
Model versions, recommendations, owner responses, outcomes and revalidation must remain linked over time.

The central gap is integrative and evaluative: whether an auditable, multi-business SME system can combine temporally valid analytics with feasible, understandable actions while retaining evidence of later outcomes. The thesis tests this bounded contribution against the supplied data and available businesses.

Table 2.1: Research gaps, responses and required evidence.

Gap	Response	Evidence required
G1 Context	Use supplied Bangladesh SME records	Manifest, source reconciliation and bounded results
G2 Integration	Versioned events and historical snapshots	Traceability from records to features, outputs and actions
G3 Validity	Chronological splits, common baselines and frozen evaluation	Out-of-time metrics, ablations, calibration and leakage checks
G4 Decision	Constraint-aware purchasing and retention logic	Policy replay using reconciled costs and constraints
G5 Explanation	Owner-readable evidence, uncertainty and source details	User-task correctness and appropriate interpretation
G6 Lifecycle	Versioning, monitoring and outcome tracking	Revalidation and recommendation-outcome records

The contribution is a reproducible process for selecting suitable methods, linking estimates to operational constraints and limiting claims to the evidence actually tested.

Chapter 3

Proposed Methodology

3.1 Multi-Sector Applicability and Validation

The intended scope is SME business analytics across sectors, rather than one industry. A common core models organizations, branches, transactions, expenses, payments, customers and suppliers. Configurable sector extensions preserve differences in products, services, inventory, resource capacity and operating rules. Demand and reorder experiments apply to inventory-based firms; service firms require service-demand and capacity tasks. Customer-risk tasks require stable pseudonymous identifiers and adequate follow-up in either setting.

Sampling and reporting will retain sector and business identifiers. Results will be reported per business and per sector before aggregation. Pooled models will be compared with sector-specific and business-adapted alternatives using the same chronological cutoffs. Held-out-business tests assess transfer within represented sectors; held-out-sector tests assess transfer across sectors when enough independent sectors and businesses are available. In their absence, cross-sector transfer remains untested. Preprocessing is fitted only on training data, and firms with many rows must not dominate business-level conclusions. No universal accuracy or business-value claim follows from framework coverage.

3.2 Research Approach

3.2.1 Research Strategy and Design-Science Rationale

This thesis adopts a design-science research approach supported by empirical evaluation. Design science is appropriate because the study develops an artifact intended to address an applied organizational problem and evaluates that artifact against explicitly defined requirements [10]. The principal artifact is the B-SMART framework, including its canonical operational data design, time-aware analytical pipeline, constraint-aware recommendation process, and mechanisms for presenting supporting evidence to the owner.

Empirical evaluation is necessary because a plausible technical architecture alone is insufficient to establish predictive validity, business usefulness, operational feasibility, or usability. Accordingly, the research evaluates not only whether the proposed system can be implemented, but also whether its analytical outputs remain temporally valid, whether its recommendations respect the operating constraints of participating businesses, and whether intended users can interpret and appropriately use those recommendations.

3.2.2 Research Stages and Units of Analysis

The work follows four stages: define the problem and evidence; design the operational schema, snapshots, models and decision rules; evaluate data integrity, prediction and robustness; and assess decision consequences and owner use. The four stages are conducted sequentially, with feedback permitted during development but not after the final test is frozen. Later findings may refine the artifact but cannot alter a frozen final test.

The unit of analysis depends on the task: a shop–category or SKU time series for forecasting, a pseudonymous customer snapshot for inactivity, and an eligible action under declared business constraints for strategy evaluation. Evidence from one unit is not treated as proof at another level.

3.2.3 Design Requirements and Acceptance Logic

The artifact is assessed against eight requirements:

DR1. **Decision relevance:** identify user, unit, horizon and decision.

DR2. **Operational integrity:** preserve identifiers, event times and corrections.

DR3. **Temporal validity:** use only information available at the cutoff.

DR4. **Proportionate modelling:** compare every candidate with a baseline.

DR5. **Uncertainty:** report intervals, calibration and sensitivity where relevant.

DR6. **Feasible action:** enforce stock, cash, lead-time, pack, expiry and consent constraints.

DR7. **Human authority:** allow owners to inspect, modify or reject actions.

DR8. **Reproducibility:** retain data, feature, model and output versions.

Acceptance is non-compensatory: accuracy cannot compensate for leakage, privacy failure or infeasible action. Requirements are classified as met, partially met, not met or not evaluable. Missing evidence reduces the strength of a conclusion rather than being replaced by an assumption.

3.2.4 Research Questions, Hypotheses, and Measures

The research questions introduced in Chapter 1 are operationalized as follows:

RQ1 / H1

A business demand model incorporating locally available calendar, price, promotion, and market-context features will reduce out-of-time WAPE or MASE relative to both an otherwise identical model without those feature groups and a seasonal-naive benchmark.

RQ2 / H2

A model trained from historically valid customer snapshots will predict future inactivity better than an event base-rate or rule-based baseline, as measured by PR-AUC, and will achieve Lift@10% greater than 1.0 without unacceptable deterioration in probability calibration.

RQ3 / H3

Under identical demand paths, costs, and operational constraints, the proposed forecast-based replenishment policy will reduce the combined consequences of stock-outs and inventory holding relative to the current replenishment policy used by the participating owner.

RQ4 / H4

In counterbalanced owner decision tasks, an explanation-supported condition will improve decision accuracy or confidence without causing a practically detrimental increase in task-completion time relative to an output-only condition.

Table 3.1: Alignment of research questions, evidence, comparisons, and primary measures.

RQ	Unit and Comparison	Primary Measures	Evidence Required
RQ1	Unseen branch–SKU periods; context-aware model versus ablated models and forecasting baselines	WAPE, MASE, interval coverage, and related out-of-time forecast measures	Time-stamped sales, stock, price, promotion, calendar, and relevant context data
RQ2	Future customer outcome windows; predictive model versus base-rate or rule-based baseline	PR-AUC, Lift@10%, Brier score, and calibration measures	Pseudonymous customer transactions with historically valid feature cutoffs and sufficient future follow-up
RQ3	Identical demand paths; proposed replenishment policy versus the participating owner's existing policy	Stock-out units, inventory holding, and total cost under declared assumptions	Stock records, lead times, purchasing constraints, and validated cost assumptions
RQ4	Equivalent decision situations; explanation-supported output versus output-only condition	Decision correctness, completion time, confidence, and perceived usability	Consenting target users and counterbalanced decision tasks

The final test set will not be inspected during hypothesis development, model selection, feature selection, or primary metric selection. Measures introduced after this stage will be identified as exploratory. Furthermore, a statistically detectable difference will not automatically be interpreted as practically important unless its magnitude, operational consequence, or cost implication is material to the target decision.

3.2.5 Target Population and Sampling Plan

The target population consists of SMEs operating in Bangladesh across retail, wholesale, distribution and service sectors. Recruitment will seek multiple independent businesses and record sector, size and operating model. An initial three-to-five-business pilot can test collection and usability, but cannot establish population-wide or cross-sector effectiveness. Broader sector comparisons require adequate independent businesses in each stratum.

The intended analytical target is approximately 12–24 months of longitudinal records across participating businesses, subject to data availability, data quality, owner consent, and ethics approval. A business with substantially less history may still contribute to testing data collection, mapping, schema, and data-quality procedures, but such a business shall not independently determine claims of temporal or cross-business generalization.

Businesses will be purposively recruited because participation depends on access to relevant longitudinal evidence, including invoice, stock, purchase, batch, and expiry information where available, together with owner agreement. At a minimum, included transaction data shall contain interpretable dates, product identifiers, quantities, and sufficient contextual information to reconstruct the relevant business process. Participating businesses shall also have an owner or responsible operator capable of describing the current replenishment process.

Customer-level analysis imposes additional eligibility requirements. It requires stable pseudonymous customer identifiers and sufficient follow-up after each historical cutoff to determine the relevant outcome. If a business, analytical task, or time period does not satisfy the minimum evidence requirements for a specific analysis, that unit will be excluded from that analysis rather than being repaired through invented identifiers, fabricated constraints, or assumed outcomes.

3.2.6 Data Sources and Evidence Tiers

Evidence will be separated into explicit tiers so that convenient external, public, or generated data cannot silently alter the claim being tested.

TABLE 3.2: Planned evidence tiers and corresponding limits on inference.

Tier	**Source**	**Authorized Use**	**Prohibited or Unsupported Inference**
1	Consenting Bangladesh business invoice, stock, purchase, batch, expiry, customer, and constraint records	Primary business forecasting, customer-risk, strategy, and usability evaluation	Generalization beyond sampled businesses or untested transfer across SME verticals
2a	Bangladesh market-price data, including relevant WFP/DAM sources [25]	Local external context and separate market-price analysis	Direct inference of SKU-level business sales, customer behaviour, or business revenue when those variables are not observed
2b	UCI Online Retail II data from the United Kingdom [3]	Real-invoice pipeline testing and portability checks for selected methods	Claims about Bangladeshi purchasing behaviour or local business effectiveness
2c	Public Bangladesh retailer demand-series data [14]	Local quantity-forecast procedure and implementation checks	Multi-SKU, customer, price, inventory, or profit claims when the required fields are absent
3	Software test records	Software testing, rare-case testing, scalability checks, and privacy-safe examples	Claims of field effectiveness, real customer behaviour, or local operational impact

Each incoming dataset will be accompanied by a manifest recording its source, owner or repository, collection period, receipt date, checksum, licence or agreement, row count, schema version, mapping rules, exclusions, and known limitations.

Tier-3 generated data will never be mixed into a Tier-1 test set. If generated or augmented records are used during model development, results from real-only training and augmented training will be reported separately and evaluated against the same untouched real test period wherever the required real evidence is available.

3.2.7 Data Acquisition and Variable Availability

The supplied package contains operational sales, customer, product, purchase, payment, inventory, return, promotion and calendar records. RQ1 uses the complete daily shop–category observed-sales panel. Other tables are joined only after their keys, event times and business meaning are checked.

Each variable is classified by availability time. Price, promotion or supplier conditions are predictors only when known at the forecast origin. Later sales, returns and inventory movements are outcomes, not predictors. Raw business files remain in controlled storage; the repository contains code, schemas, checksums and aggregate results.

3.2.8 Data-Quality Assessment and Task Eligibility

The ingestion audit validates required fields, types, identifiers, duplicate keys, date ranges, non-negative quantities and cross-table relationships. Sales lines are reconciled to parent orders; inventory sale movements are checked against sales lines; and daily aggregates are compared with transaction-level reconstruction. Mismatches are reported and quarantined, not silently repaired.

For the complete shop–category panel, the audit checks unique shop–category–date keys, required values, consistency of the zero-sales flag and a complete daily calendar. This panel is eligible for observed-sales forecasting at its declared aggregation level. It does not automatically make sparse SKU exports eligible for zero-demand labels.

SKU exposure requires evidence that the product was active, the shop was operating and stock was available. Missing SKU dates are not automatically zero. Negative reconstructed balances, incomplete inventory coverage, unknown assortment, closed periods and stock-out periods remain censored or unknown until reconciled. RQ1 observed-sales forecasting can proceed on the complete panel; SKU latent-demand estimation, customer inactivity and inventory-policy replay each require their own validated inputs.

3.2.9 Ethics, Privacy, and Governance

University ethics approval and an appropriate signed data-sharing agreement will be obtained before primary data collection begins. Direct identifiers such as customer names, phone numbers, email addresses, national identifiers, account numbers, or mobile-wallet numbers are not required for the research analytical tables.

Where customer-level analysis is permitted, each participating business will use a separate salted pseudonymous customer key. Any mapping capable of re-identification will remain under the control of the participating business and will not be stored in the research repository.

The data-sharing agreement will specify the research purpose, authorized researchers, storage arrangements, retention period, publication format, withdrawal conditions, and deletion procedures. Access to research data will follow least-privilege principles, while published outputs will consist of aggregated results or individually reviewed de-identified examples.

A model-generated customer-retention recommendation does not itself authorize contact with that customer. Customer consent and the participating business's contact policy

remain independent feasibility constraints. Similarly, system outputs are intended to support rather than replace the decision-making authority of the owner.

3.2.10 Canonical Operational and Analytical Model

The operational model links businesses and shops to products, customers, suppliers, sales orders, sales lines, payments, purchase orders, inventory movements, returns and promotions. Stable identifiers preserve relationships. Analytical tables are derived at a declared grain and cutoff; they are not independent source-of-truth records.

The implemented RQ1 grain is shop–category–day. Its target is observed units sold. SKU-level demand requires separate assortment and stock-exposure validation before missing dates can be interpreted.

3.3 Formal Problem Formulation

Let b denote a business, l a branch, p a product, r an inventory batch, c a customer, and t a decision time. Let

$$D_t = \left\{ \begin{array}{c} \text{Sales, Stock, Batches, Expiry, Purchases, Expenses,} \\ \text{Customers, Constraints, and Context} \end{array} \right\}_{\leq t}, \tag{3.1}$$

denote the governed evidence available at or before decision time t. Accordingly, any analytical output produced at t must be derived only from information contained in D_t.

3.3.1 Performance Evaluation

For each KPI K, the system returns $K(D_t; W, g)$, where W is the reporting window and g is the aggregation level. Each KPI retains its numerator, denominator, filters, definition version, and data-quality status so that the owner can reproduce and inspect the calculation.

For reporting window W, net sales are calculated from valid invoice lines using the documented business treatment of discounts, cancellations, and returns. A general form is

$$\text{NetSales}_W = \sum_{i \in W} q_i p_i - \sum_{i \in W} d_i - \sum_{r \in W} v_r, \tag{3.2}$$

where $q_i p_i$ is the gross value of invoice line i, d_i is the recognized discount, and v_r is the value of a valid return or reversal attributed to the reporting window.

Gross margin is reported only when sufficiently reliable cost evidence is available:

$$\text{GrossMarginRate}_W = \frac{\text{NetSales}_W - \text{COGS}_W}{\text{NetSales}_W}, \tag{3.3}$$

provided that $\text{NetSales}_W \neq 0$.

If cost information is unavailable for a material share of sold units, the system reports sales and cost coverage rather than silently interpreting revenue as profit. Average order value uses eligible completed orders as its denominator; repeat-rate reporting states customer-identification coverage; and stock-turnover reporting specifies both the cost and average-inventory conventions used. Period comparisons use comparable windows or explicitly identify differences in trading days and closures.

Each KPI response therefore contains a value, definition version, reporting period, aggregation level, coverage indicators, and reconciliation status. This structure makes changes in business-rule definitions visible and allows the evaluation to distinguish calculation anomalies from genuine changes in business performance.

3.3.2 Demand Prediction

The implemented RQ1 target is observed units at shop–category–day level. For series s, origin t and horizon $h \in \{1, 7, 14, 28\}$, the direct cumulative target is

$$Y_{s,t,h} = \sum_{j=0}^{h-1} y_{s,t+j}.$$

Separate models are fitted for each horizon. Features must be available strictly before t, and every training target window must end before its validation origin.

Historical features include lags 1, 7, 14, 28 and 56; rolling mean, standard deviation, sum and zero ratio over 7, 14, 28 and 56 days; days since last positive sale; ADI; positive-demand mean and CV^2; shop and category identifiers; and calendar encodings. The context-aware challenger adds past shop, category and network aggregates, recent trends and the series' share of those aggregates. Context features use only observations before the origin.

Candidates include seasonal-naive, Croston–SBA, TSB, LightGBM Tweedie, CatBoost Tweedie and a two-stage occurrence–quantity model. ADI and CV^2 support diagnosis but do not automatically route a model. Specialist methods are promoted only with adequate common-origin validation and independent series support.

The forecast estimates observed sales, not unconstrained latent demand. Stock-out censoring, promotion effects and SKU-level forecasts require additional point-in-time evidence and separate validation.

3.3.3 Adaptive Forecasting Architecture and Promotion Gates

The architecture separates source eligibility, target construction, candidate forecasts, model selection and business action. Source and target gates distinguish observed zero from censored or unknown demand. Candidates are compared on common rolling origins, bias and independent-series support. Uncertainty and aggregation follow selection; inventory actions require validated costs, lead time, stock and owner constraints.

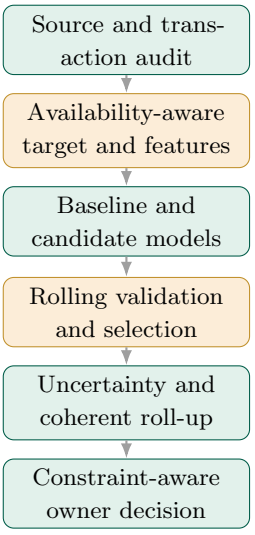


FIGURE 3.1: Evidence-gated adaptive demand-forecasting architecture.

3.3.4 Future Customer Inactivity

For customer c at historical cutoff t, future inactivity is represented as

$$q_{c,t} = P(\text{no purchase in } (t, t+H] \mid X_{c,t}), \tag{3.4}$$

where $X_{c,t}$ contains only records available by cutoff t, and H is the predefined future outcome horizon. The value of H will be fixed before final evaluation with reference to the relevant business purchase cycle.

The resulting $q_{c,t}$ represents predicted inactivity risk and does not, by itself, estimate the causal effect of contacting customer c.

3.3.5 Strategy Optimization

For an eligible action $a \in A_t$, the system evaluates

$$U(a \mid D_t) = \mathbb{E}[\Delta\text{Profit}(a)] - \text{ActionCost}(a) - \lambda\,\text{Risk}(a), \tag{3.5}$$

subject to operational constraints including budget, storage, minimum-order quantity, supplier lead time, pack multiple, remaining shelf life, expiry, temperature-controlled storage capacity, incoming stock, available inventory, and customer-contact consent where applicable.

This transparent objective represents the transition from predictive outputs to prescriptive decisions. Its terms will be estimated using available evidence, and uncertain values will be examined through sensitivity analysis rather than treated as precisely known constants.

3.3.5.1 Inventory Position and Replenishment Cost

Inventory position combines usable on-hand stock and confirmed incoming quantity, less committed demand. Replenishment is constrained by supplier lead time, pack multiple, minimum order quantity, budget, storage, expiry and remaining shelf life. An action is withheld when a required field is unknown or a hard constraint is violated.

Policy evaluation compares the owner policy and proposed policy on the same demand path and opening inventory. Outcomes include stock-out units, average inventory, holding cost, shortage cost, expiry and total cost. Forecast error alone is not interpreted as inventory benefit.

3.3.5.2 Customer Action Logic and Abstention

Customer-action logic differs from replenishment logic. The inactivity probability $q_{c,t}$ in Equation 3.4 may be used to prioritize customers for review, but it does not estimate the causal effect of contacting customer c.

A contact action is eligible only when customer consent and business policy permit contact, the proposed treatment is specified, and sufficient operational capacity exists. Expected profit will be calculated only when a defensible estimate of incremental response benefit is available. Otherwise, the output will explicitly remain a risk-ranked contact list. This separation prevents a predictive probability from being incorrectly interpreted as campaign uplift.

The action layer may also abstain from making a recommendation. Abstention is appropriate when required constraints are missing, uncertainty makes the leading alternatives practically indistinguishable, predicted benefit does not exceed action cost over the declared sensitivity range, or the relevant model fails a monitoring gate. An abstention accompanied by a reason is therefore treated as valid safety behaviour rather than as a system error.

3.3.6 Preprocessing and Historical Snapshots

The processing sequence is fixed: validate the source and schema; remove or flag exact duplicates; reconcile cancellations and returns; construct the data split; fit imputation,

encoding, and scaling transformations using training data only; construct features; train candidate models; and evaluate them on the untouched test set. This ordering prevents information from future periods from leaking into model development [15].

Demand features may include lagged demand, rolling location and variability statistics, day of week, month, distance to relevant festivals, price, discount, known promotions, stock-out indicators, pack units, remaining shelf life, and external market context. Batch-expiry and temperature-controlled storage variables primarily constrain action feasibility and enter prediction only when their decision-time availability and analytical role are defensible.

Where feasible, zero observed sales caused by zero demand will be distinguished from zero sales caused by lack of stock. Intermittency will be measured before selecting temporal granularity and candidate forecasting methods.

Customer snapshots may include RFM variables, tenure, order-value statistics, category breadth, purchase intervals, discount behaviour, and pre-cutoff return rates. The outcome label is constructed from a subsequent future window. Where required, a purge interval will separate the observation and label windows to prevent overlap or delayed events from introducing leakage.

Customer identifiers used for an entity-generalization test will not appear on both sides of the corresponding training–test boundary.

3.4 Candidate Methods and Selection Rules

3.4.1 Baseline Ladder and Task-Specific Candidates

Method selection follows a baseline ladder, with model eligibility determined by the available evidence volume and task structure.

- **Performance:** Auditable KPI definitions and rule-based anomaly detection provide the descriptive baseline.
- **Demand:** Last-value, weekly seasonal-naive, moving-average, and exponential-smoothing baselines are evaluated before ARIMA/SARIMA, Croston–SBA, TSB, and gradient-boosted lag models. LightGBM and CatBoost are candidates for pooled shop/category or SKU features when the sample and categorical structure support them. LSTM models are considered only when sequence volume and history justify their additional complexity. Intermittent-demand methods are promoted only after same-origin validation [6, 24, 26, 27].
- **Inactivity:** A recency or base-rate rule is compared with Logistic Regression, Random Forest [2], and XGBoost [5]. Class weighting and decision thresholds are selected using development data only.

- **Segmentation:** Appropriately scaled RFM features are analysed using K-Means. Candidate values of K are evaluated using inertia, silhouette, resampling stability, and business interpretability.
- **Explanation:** Coefficient- or rule-based explanations are preferred for simple models, while SHAP provides model-behaviour attributions for compatible tree-based models [16]. Displayed explanations will remain selective and context-specific [18].

3.4.2 Model Eligibility and Sector-Aware SME Adaptation

Candidates are compared only when target, origin, horizon, training information and evaluation rows are aligned. Every task begins with a baseline. Pooled models may borrow information across businesses; business adaptation requires sufficient independent history.

Selection considers error, aggregate bias, calibration, stability across origins and independent-series support. Repeated forecasts of one series do not create independent businesses. Regime labels support diagnosis but do not justify automatic specialist routing when subgroup support is sparse. Sector-specific extensions are enabled only when relevant fields and operating rules are available.

3.5 The B-SMART Sequential Procedure

3.5.1 Strategy Recommendation Algorithm

Algorithm 1: B-SMART Strategy Recommendation

Input: Data available at cutoff D_t, horizon H, constraints C_t, validated model/baseline versions M, and candidate actions A

Output: Ranked feasible recommendations R_t

Check sources, schema, timestamps, and organizational ownership;
Create the historical feature snapshot X_t;
Apply transformations fitted on the appropriate training evidence;
Compare eligible baseline, pooled-business, and shop-adapted candidates;
Estimate demand uncertainty and customer inactivity probabilities;
Compute reproducible KPIs and data-quality alerts;
Generate candidate actions with source and decision-time metadata;
foreach $a \in A$ **do**
    Check budget, stock, pack, applicable shelf-life, storage, lead-time, capacity, and consent feasibility under C_t;
    Estimate benefit, cost, and uncertainty using declared assumptions;
    $U(a) \leftarrow \mathbb{E}[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\,\text{Risk}(a)$;
Remove infeasible actions and rank the remaining actions by $U(a)$;
Attach a concise explanation, uncertainty statement, cutoff, and evidence source;
Return the top-K actions and record the owner's response and any subsequent observable outcome;

3.5.2 Computational Complexity

For N invoice lines, d validated fields, and P candidate actions, source validation and feature aggregation are approximately

$$O(Nd), \tag{3.6}$$

while ranking the candidate actions requires approximately

$$O(P \log P). \tag{3.7}$$

The computational complexity of model training depends on the selected model and will be measured empirically. Serving uses versioned artifacts and therefore does not retrain the model within the request path.

3.6 Proposed Architecture and Information Flow

3.6.1 Architecture Components

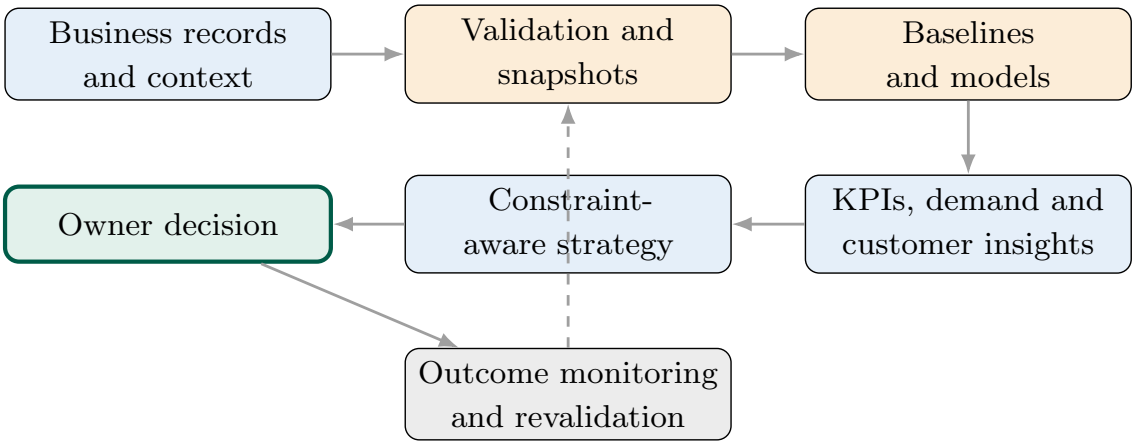


FIGURE 3.2: System information flow and owner oversight.

The operational layer preserves business boundaries and source lineage. The analytical layer compares baselines and eligible models, then produces KPIs, forecasts and customer insights. The strategy layer combines those outputs with business constraints before displaying a recommendation. Owners remain decision-makers. Outcomes enter monitoring and become training evidence only after reconciliation and validation for a later model version.

3.6.2 Monitoring and Continuous Learning

Information flow is deliberately one-way at prediction time: future outcomes cannot alter the historical snapshot that generated an earlier prediction or recommendation.

Operational state and recommendations may be updated when relevant events occur or according to a defined schedule. However, the deployed model is not blindly retrained after every transaction.

Observed outcomes enter the monitoring process and become eligible training evidence only in a later model version after reconciliation, label maturation, drift assessment, and comparison with the existing model and baseline.

This monitored feedback lifecycle represents the intended meaning of continuous learning while preserving the ability to reproduce what the system knew when a particular action was proposed.

3.7 Snapshot Lineage and Frozen Evaluation Flow

3.7.1 Historical Snapshot Construction

Each snapshot is defined by cutoff t. Only records available by that cutoff may enter the prediction. Event time and recording time are retained separately so late-arriving or corrected records cannot silently change a past information set.

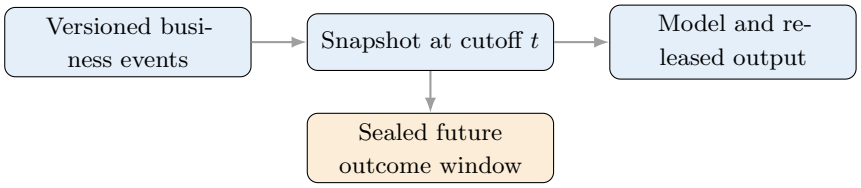


FIGURE 3.3: Point-in-time snapshot and future-outcome separation.

For forecasting, each training target window ends before its validation origin, and transformations are fitted within the training window. Customer labels are assigned only when the complete future horizon is observable. Overlapping outcomes require an appropriate purge interval.

3.7.2 Frozen Final Evaluation and Outcome Maturation

Training and validation snapshots are used for model development. The final test set remains sealed until the task definition, feature and eligibility rules, candidate models, primary metrics, decision thresholds, and random seeds have been frozen.

If a defect is identified after the final test has been opened, the defect and its impact will be documented. The affected observations cannot subsequently be treated as though they were still part of an untouched final test. This rule protects the interpretation of the final result rather than merely enforcing the mechanics of a data split.

Outcome records are not automatically incorporated into a released model. Observed outcomes first enter the monitoring store. They become eligible for a later model version only after reconciliation, label maturation, and repetition of the appropriate validation protocol.

This delay ensures that an owner's response, a corrected invoice, or a partially observed outcome cannot retrospectively change the evidence that supported a historical production decision.

3.8 Owner-Facing Recommendation Evidence Contract

3.8.1 Minimum Recommendation Evidence

Each recommendation identifies its business, decision, target, cutoff, model or rule version, uncertainty and supporting evidence. Reorder suggestions expose current and incoming stock, forecast horizon, lead time, pack requirements, budget and expiry risk. Customer actions expose the historical behaviour supporting the priority, probability reliability and consent status.

When a required field is missing, the system withholds the action or labels the result incomplete. It does not assume costs, stock or customer responses. Owners may accept, reject, modify or defer actions; responses and later observable outcomes are retained for audit.

3.8.2 Human Use and Accountability

An explanation is considered successful only when it contributes to correct and appropriate use. The owner study will therefore include scenarios in which the appropriate decision is to reject or defer an apparently attractive recommendation.

This evaluates calibrated human reliance rather than merely whether participants prefer a more detailed interface. The evidence contract also supports failure analysis because an inappropriate decision can be traced to the underlying estimate, missing constraint, explanation, outdated evidence, or owner interpretation rather than being attributed generically to the "AI."

3.9 Evaluation Protocol

Evaluation follows development, protocol freeze, untouched testing and bounded reporting. Candidates, features, thresholds and primary metrics are selected using development data only. The final test is opened after the protocol is frozen; later changes are exploratory.



FIGURE 3.4: Frozen evaluation sequence.

3.9.1 Forecasting Evaluation

All forecasting splits preserve temporal order. Candidate methods are selected using rolling-origin or expanding-window validation, followed by one evaluation on a final untouched period.

For observed values y_i and forecasts $\hat{y}_i$, the principal error measures include

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|\,, \tag{3.8}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{3.9}$$

$$\mathrm{WAPE} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\sum_{i=1}^{n} |y_i|}, \qquad \sum_{i=1}^{n} |y_i| > 0. \tag{3.10}$$

MASE will additionally support scale-independent comparison and avoids several limitations of conventional percentage errors [13]. For the current cumulative-horizon benchmark, the reported horizon-scaled weekly MASE uses the per-series mean absolute lag-7 difference multiplied by the forecast horizon. Cluster bootstrap resampling is performed at the shop–category-series level to preserve repeated-origin structure. Prediction-interval coverage and interval width will be used to evaluate uncertainty.

Results will be stratified by demand volume and intermittency so that poor performance on low-volume or sparse products is not hidden inside an aggregate average.

3.9.2 Customer and Segmentation Evaluation

Customer-inactivity models will report outcome prevalence, PR-AUC, ROC-AUC, Lift@10%, Brier score, and probability-reliability analysis. Precision–recall measures are emphasized under class imbalance [21], while calibration is evaluated independently because a model may produce useful rankings while still producing misleading probabilities [19].

Decision thresholds and contact fractions will be selected using validation evidence and declared cost assumptions. They will not be modified after the final test outcomes have been inspected.

Clustering evaluation will report inertia, silhouette, cluster stability, minimum cluster size, and corresponding RFM profiles. Final segment interpretations will also be reviewed with domain participants.

If separation is weak or unstable, the result will be reported as such. The existence of algorithmically generated clusters will not be treated as evidence that natural or causal customer types exist.

3.9.3 Ablation, Robustness, and Statistical Analysis

Planned ablations include local-calendar features, price and promotion features, market context, pack and expiry context, and customer-history extensions.

Comparisons will include the transparent baseline, eligible pooled-business models, and shop-adapted models where the available evidence permits them. Where augmentation is used, real-only and augmented variants will be evaluated under controlled comparisons.

Results will be presented by business, with paired comparisons conducted over common forecast origins or comparable analytical instances. A held-out-business transfer test will be used where sufficient participating businesses are available.

Uncertainty will be summarized using paired block-bootstrap intervals across time or business units, depending on the relevant dependence structure. Differences between models will be reported using effect estimates and uncertainty intervals rather than significance labels alone.

Sensitivity analysis will vary supplier lead time, shortage cost, holding cost, action cost, and the action-risk penalty used in Equation 3.5.

3.9.4 Frozen Analysis and Statistical Decision Rules

One primary comparison and metric per research question are frozen before the final test. RQ1 uses paired out-of-time errors on common origins; RQ2 uses PR-AUC with calibration as a separate criterion; RQ3 compares policies on identical demand paths and constraints; and RQ4 evaluates task correctness under counterbalanced owner tasks.

Uncertainty methods must respect dependence: neighbouring dates, repeated customer snapshots and observations from one business are not independent. Paired block-bootstrap or cluster methods are selected according to the unit of analysis. Effects are reported with intervals and interpreted for practical importance, not significance labels alone.

3.9.5 Practical Importance and Missing Outcomes

Practical importance will be evaluated independently of statistical evidence. A small reduction in average forecasting error may be operationally important for a high-volume product but irrelevant after ordering constraints are applied. Similarly, a moderate increase in task-completion time may be acceptable if it produces a material improvement in decision correctness.

Where defensible cost or effect thresholds can be established with participating businesses, they will be specified before the final comparison. Where no defensible threshold exists, the magnitude and uncertainty of the effect will be reported without assigning an arbitrary success label.

Secondary metrics, alternative horizons, customer segments, and feature interactions remain useful for diagnosis but will be identified as secondary or exploratory analyses. When many related comparisons are examined, the full set of relevant results will be reported rather than only favourable cases.

Any post-test change to a model, threshold, feature definition, or analysis creates a new exploratory version rather than replacing the original primary result. This separation allows continued model improvement without rewriting the original evidence.

Missing outcomes will be explicitly identified. An incomplete future horizon cannot automatically be treated as evidence of customer inactivity. An unknown inventory balance cannot be treated as zero stock. Similarly, a recommendation without a later observable outcome may support a usability analysis but cannot support a claim of realized business impact.

Where assumptions can reasonably be bounded, sensitivity analysis will be used. Otherwise, the corresponding claim will be classified as not evaluable with the available evidence.

3.9.6 Decision, Usability, and Scalability Evaluation

Replenishment policies are compared on identical demand paths and opening inventory using stock-out units, average inventory, holding cost, shortage cost, expiry and total cost. Retention value is estimated only when incremental response, contact cost and consent evidence exist; otherwise the task remains prioritisation, not realized-profit estimation.

Owner evaluation measures decision correctness, completion time, confidence, error count and usability. Generated test records may support software-load and scalability checks, but are not mixed with empirical forecast or business-impact results.

3.9.7 Monitoring Plan

Monitoring will distinguish among schema or data-quality failure, input distribution shift, calibration change, and outcome-performance deterioration.

An alert does not automatically authorize retraining. Any replacement model must pass the validation procedure again and retain its training window, feature version, parameters, and comparison results.

This requirement ensures that adaptation is evaluated before deployment rather than assumed to be beneficial merely because new data have become available.

3.10 Reproducibility, Audit, and Failure Reporting

3.10.1 Reproducibility Levels

Four levels of reproducibility will be designed.

- **Data reproducibility:** requires a source manifest, checksum, schema version, mapping rules, and exclusion log.
- **Analytical reproducibility:** requires snapshot identifiers, fitted transformations, feature definitions, target construction, and split manifests.
- **Model reproducibility:** requires code, execution environment, parameters, random seeds, trained artifacts, and the evaluation command.
- **Decision reproducibility:** additionally requires the applicable constraints, cost assumptions, candidate-action set, ranking rule, and interface version.

3.10.2 Restricted and Shareable Evidence

Primary partner data do not need to be publicly released in order for the study to maintain reproducibility controls. The reproducibility package will therefore distinguish restricted evidence from shareable artifacts.

Shareable materials may include schemas, data dictionaries, manifest templates, transformation code, generated test fixtures, and configuration files without revealing confidential identifiers or transaction records.

Where restricted evidence is available to an authorized evaluator, the available package should permit that evaluator to reconstruct the relevant snapshot and result. An external reader should additionally be able to inspect the process using a public or generated test fixture.

3.10.3 Experiment Records and Correction History

Each experiment will generate a machine-readable run record containing, where applicable, the research question, dataset and snapshot hashes, training and test windows, grouping keys, feature version, candidate and baseline versions, parameters, output-artifact hashes, warnings, and start and finish times.

Manual spreadsheet corrections will not be made outside this record. If an owner supplies a correction, it becomes a dated source or mapping revision. The correction affects downstream outputs only when a new version is executed.

3.10.4 Failure Reporting

Failure reporting is part of the scientific output. Data-eligibility failure, lack of convergence, numerical instability, training failure, resource exhaustion, and constraint infeasibility will be distinguished from ordinary evaluation outcomes.

The number of attempted cases will be reported. Results will not be summarized only over successful model runs while silently excluding failures.

A method that fails on sparse or short series will therefore be evaluated against the full eligible set to which it was applied, rather than only the subset for which it returned a prediction.

3.10.5 Recommendation Auditability and Claim Traceability

Auditability extends to the owner interface. Every recommendation receives an identifier linking the evidence contract, displayed recommendation, owner response, and any later observable outcome.

If a recommendation changes following a data correction or model replacement, the previous record remains immutable and the successor record refers to it. This supports accountability without implying that the model, researcher, or owner alone caused the eventual business outcome.

The final research package will also include a claim-to-evidence index. Each central claim will identify the corresponding research question, figure or table, experimental record, and evidence tier.

Claims supported only by public, generated, simulated, or small-sample evidence will remain explicitly labelled as such. This index provides a final safeguard against expanding the scope of a claim beyond what was actually evaluated.

3.11 Threats to Validity and Planned Controls

The main threats are source-field errors, incomplete inventory exposure, business heterogeneity, temporal leakage, sparse demand, uncertain costs and limited independent businesses. Controls include transaction reconciliation, task eligibility, point-in-time features, common-origin baselines, cluster-aware uncertainty, sensitivity analysis and an untouched final test.

The supplied records support the current shop–category observed-sales experiment. They do not alone establish SKU latent demand, causal promotion effects, inventory-policy savings or transfer to every SME sector. Those claims require additional validated fields or separate participant-based evaluation. Unavailable evidence is reported as a limitation.

3.12 Expected Methodological Outputs

At completion, the methodology is expected to produce a documented canonical business schema and data dictionary, a versioned feature and partner-data protocol, and explicit specifications for transparent baselines, pooled-business models, and eligible shop-adapted models.

The study will also produce a reproducible evaluation package, a constraint-aware recommendation procedure, and an owner-study instrument.

Effectiveness claims will be restricted to the sampled businesses, sectors, tasks and periods actually evaluated. Transfer to a new sector requires schema and constraint mapping, new evidence and held-out-sector evaluation.

The system supports business decisions under owner oversight; automated actions outside the evaluated business tasks remain excluded.

Implementation details, observed numerical results, and conclusions will be reported only when the corresponding empirical evidence is available.

Chapter 4

Implementation

4.1 Dataset Package and Research Scope

The supplied dataset manifest identifies the package as 100% real SME sales and real people, with Bangladesh as the business context, BDT currency and coverage from 1 January 2023 to 31 December 2025. It lists 32 shops, 11,000 customers, 480 products and 56 suppliers. The source package includes 365,447 sales orders, 736,683 sales lines, 381,756 payments, 122,422 purchase orders and purchase lines, 994,808 inventory movements, 134,379 inventory lots, 5,102 returns, 1,216 promotions and 1,096 calendar rows.

The RQ1 experiment uses the dedicated complete daily shop–category panel: 32 shops, 16 categories, 512 series and 1,096 daily observations per series, for 561,152 rows. The panel contains 1,580,361 observed units. The separate simulated unmet-demand table is not used as observed sales or as empirical lost-sales ground truth. Raw business records remain in controlled storage; the public repository contains code, schemas, checksums and aggregate outputs.

TABLE 4.1: Source-table coverage declared in the supplied real-business dataset manifest.

Source table	Rows	Analytical role	Use in current RQ1
Shops	32	Business and shop identifiers	Series grouping
Customers	11,000	Customer-level operational history	Not used as a forecast target
Products	480	SKU and category master	Category mapping and reconciliation
Suppliers	56	Supplier and procurement context	Not yet used as a predictor
Sales orders	365,447	Order headers and shop/customer links	Parent transaction validation
Sales lines	736,683	SKU-level quantities and values	Source reconciliation and aggregation
Payments	381,756	Payment events and settlement evidence	Not used as a forecast predictor
Purchase orders / lines	122,422 / 122,422	Procurement and received quantities	Inventory reconciliation
Inventory movements	994,808	Opening, purchase, sale and adjustment events	Stock-exposure diagnostics
Inventory lots	134,379	Lot and batch traceability	SKU eligibility diagnostics
Returns	5,102	Reversal and return events	Reconciliation context
Promotions	1,216	Promotion records	Not used until point-in-time approval is validated
Calendar	1,096	Daily calendar coverage	Calendar feature validation

The manifest declares the source package to contain real SME sales and real people. The table distinguishes source coverage from the fields actually used by RQ1: the presence of a source table does not make every field an eligible predictor. A variable is admitted only after its join key, business meaning and availability time have been checked.

4.2 Data Validation and Preprocessing

The ingestion procedure parses the date field, checks required columns and missing values, rejects duplicate shop–category–date keys, verifies that units sold are finite and non-negative, and checks that the zero-sales flag agrees with the observed quantity. It then verifies that every shop–category series contains the same complete daily calendar. No missing dates are filled by assumption. The audit found zero duplicate keys, zero missing required values, zero zero-flag mismatches, and zero missing dates in the supplied panel. The zero-sales share is 37.5962%.

The dedicated SKU-level export is not substituted for this complete shop–category panel. Its absent SKU-date rows require a separate source-aware decision about assortment, shop opening, product activation, stock-outs, and export completeness before zero demand can be assigned. This distinction prevents the conversion of unobserved records into artificial zero labels.

Table 4.2: Automated integrity checks on the complete shop–category panel.

Check	**Observed result**	**Interpretation**
Panel rows	561,152	512 series multiplied by 1,096 dates
Duplicate shop–category–date keys	0	Key uniqueness passed
Missing required values	0	Required panel fields complete
Zero-flag / quantity mismatches	0	Boolean zero indicator agrees with quantity
Missing daily dates within series	0	Complete daily panel; no date imputation required
Observed units	1,580,361	Total target quantity in the panel
Explicit zero-sales rows	210,972	37.5962% of panel observations

The zero-sales flag is checked against the observed quantity rather than trusted as an independent label. A row is an observed zero only because the complete shop–category panel explicitly contains that date and quantity. This rule must not be transferred automatically to the sparse SKU export, where an absent row may mean an inactive product, stock-out, closed shop, incomplete export or genuinely unobserved period.

4.3 Transaction and Panel Reconciliation

The 736,683 sales lines join to 365,447 sales orders without missing order references or shop mismatches. Aggregating source sales lines by `shop_id`, `sku_id` and calendar date reproduces all 699,220 rows of the prepared positive-sale SKU table exactly, including 1,580,361 units and the recorded sales value to floating-point precision. This is an internal reconciliation of the supplied real-business records; it confirms record-level consistency across the linked tables.

The prepared SKU export contains 32 shops, 480 SKUs and 13,271 shop–SKU combinations with at least one positive-sale record. It contains only positive-sale days; its `zero_day_interpretation` field warns that missing SKU dates are not automatically zero demand. The complete shop–category panel contains five fields: `shop_id`, `category`, `date`, `units_sold_observed`, and `is_zero_observed_sales`. It has 561,152 rows, 32 shops, 16 categories, 512 series and 1,096 consecutive daily observations per series. The total is 1,580,361 units and 210,972 rows (37.5962%) are zero-sales observations. Aggregating SKU sales to shop–category–date reproduces all 350,180 positive category-day rows exactly; the remaining 210,972 category rows are explicit zero-sales rows in the complete panel.

Table 4.3: Cross-table reconciliation results and unresolved exceptions.

Reconciliation	Result	Research implication
Sales lines to order headers	736,683 lines join to 365,447 orders with no missing order reference or shop mismatch	Parent transaction linkage passed
SKU daily aggregation	699,220 positive-sale rows and 1,580,361 units reproduced from source lines	Prepared SKU aggregate is internally reproducible
Shop–category aggregation	350,180 positive category-days reproduced; remaining 210,972 rows are explicit zero observations	Complete category panel reconciles to SKU sales
Sale inventory movements	All 736,683 match sales lines by order, shop, SKU, lot and quantity	Sale ledger linkage passed
Purchase movements	121,107 movement groups match received purchase-order/SKU groups	Received procurement quantities reconcile
Pending purchase lines	1,315 have no final received quantity at cutoff	Remain pending; not counted as available stock

4.4 Inventory and Procurement Reconciliation

The inventory ledger contains 13,272 opening movements, 121,107 purchase movements, 736,683 sale movements, 118,673 expiry-disposal movements and 5,073 damage movements. Every sale movement matches a sales line by order, shop, SKU, lot and quantity. The 121,107 purchase movement groups match received purchase-order/SKU quantities; 1,315 purchase lines have no final received quantity and remain pending at cutoff.

A chronological stock reconstruction identifies 269 negative-balance movement events across 253 shop–SKU combinations. It also identifies 330 positive-sale days with non-positive reconstructed intraday stock. These records are quarantined from SKU exposure labels pending correction. Across the 13,272 inventory-tracked shop–SKU combinations and 1,096 dates, the ledger reconstructs 13,781,575 fully in-stock days and 764,537 days with non-positive intraday balance. These are derived diagnostics, not an independent stock audit. Every zero-sales shop–category day has at least one recorded SKU with positive reconstructed stock during that day; this does not rule out item-level stock-outs or unrecorded assortment.

Table 4.4: Inventory movement composition and reconstructed balance diagnostics.

Movement type	**Rows**	**Treatment**
Opening	13,272	Initial recorded stock position
Purchase	121,107	Reconciled to received purchase-order/SKU groups
Sale	736,683	Matched to sales lines
Expiry disposal	118,673	Removed from usable inventory
Damage	5,073	Removed from usable inventory
Total	**994,808**	Ledger movement count

Table 4.5: Stock reconstruction exceptions requiring quarantine before SKU target construction.

Diagnostic	**Count**	**Interpretation**
Negative-balance movement events	269	Across 253 shop–SKU combinations; investigate ledger sequence or opening balance
Positive-sale days with non-positive intraday minimum stock	330	Cannot be treated as fully in-stock exposure without correction
Reconstructed fully in-stock days	13,781,575	Derived ledger diagnostic across 13,272 tracked shop–SKU combinations
Days with non-positive intraday balance	764,537	Exposure is not eligible as a known positive-stock day

The reconstruction is a derived diagnostic, not an independent physical stock count. Every zero-sales shop–category day has at least one recorded SKU with positive reconstructed stock during that day, but this does not rule out item-level stock-outs or unrecorded assortment. The 2,088 theoretical shop–SKU combinations without an opening movement are not assumed to be active series. These limitations justify keeping the current RQ1 target at shop–category observed sales until SKU exposure is validated.

4.5 RQ1 Forecasting Pipeline

For each shop–category series, the feature builder uses only information strictly before the forecast origin. Features include lags of 1, 7, 14, 28, and 56 days; rolling mean, standard deviation, sum, and zero-sales ratio over 7, 14, 28, and 56 days; days since the last positive sale; ADI; positive demand mean and CV^2; shop and category identifiers; weekday, day, month, ISO week, and cyclical calendar encodings. No future observed sales are used as predictors.

Four direct cumulative targets are evaluated: 1, 7, 14, and 28 days. Development forecast origins are 1 July, 1 August, and 1 September 2025. The previously inspected fourth-quarter 2025 period is not used for model selection. Historical training examples are

sampled every 28 days; each training label window must finish before the corresponding validation origin. A 56-day history is required before a feature row is admitted.

The LightGBM Tweedie candidate uses variance power 1.3, 100 boosting iterations, learning rate 0.035, 31 leaves, minimum child samples 80, row and column subsampling 0.85, subsampling frequency 1, L2 regularisation 2.0 and seed 42. The CatBoost Tweedie challenger uses 120 iterations, depth 6, learning rate 0.04 and L2 regularisation 5. The two-stage CatBoost hurdle uses a 100-iteration occurrence classifier and a 100-iteration positive-quantity regressor, depth 5 and learning rate 0.04; the classifier uses balanced class weights. Its probabilities are not calibrated and its output is not selected for deployment. Croston–SBA and TSB use $\alpha = 0.1$ (and $\beta = 0.1$ for TSB). The weekly seasonal-naive baseline averages up to four previous observations for each matching weekday and sums them across the horizon. All methods are evaluated on the same 512 series at each of the three origins, giving 1,536 series-origin forecasts per horizon.

Table 4.6: Feature groups and point-in-time availability rules for the RQ1 model.

Feature group	**Variables**	**Availability rule**
Own-series lags	1, 7, 14, 28 and 56 days	Observations strictly before forecast origin
Rolling demand	Mean, standard deviation, sum and zero ratio over 7, 14, 28 and 56 days	Window ends at origin minus one day
Intermittency	Days since last sale, ADI, positive-demand mean and CV^2	Calculated from pre-origin history only
Calendar	Weekday, day, month, ISO week and cyclical encodings	Calendar known at the forecast origin
Cross-series context	Shop, category and network totals; trends; series shares	Aggregates use only pre-origin observations
Potential business covariates	Price, discount, promotion, supplier and stock context	Excluded unless point-in-time availability and joins are validated

The feature policy prevents target leakage at two levels. First, no feature window includes the date being forecast or any later date. Second, a historical training example is admitted only when its entire cumulative target window ends before the validation origin. The same origin, target definition and eligible rows are used for baseline and context comparisons.

4.6 Context-Aware Forecasting Model Development

The existing pooled LightGBM Tweedie model was retained as the controlled baseline. The challenger used the same estimator family and settings while adding cross-series information available before each forecast origin: shop-level sums and means, category-level sums and means, network totals, recent shop/category trends and the focal series' share of its shop, category and network totals. This tests whether related series provide useful context without introducing future information.

The seven-day comparison uses nine monthly origins from January to September 2025, with all 512 series evaluated at every origin. One-, 14- and 28-day comparisons use three origins from July to September 2025. Q4 2025, which had already been inspected, is not used for model selection. Every training target window ends before its validation origin. Baseline and context models use the same target, origins and LightGBM settings.

The model remains a shop–category observed-sales forecaster. It does not convert missing SKU dates to zero or claim to estimate unconstrained latent demand. Code and aggregate outputs are stored under `research/forecasting/context_aware_experiment.py` and `research/forecasting/results/context_aware_v1/`.

Table 4.7: LightGBM Tweedie configuration shared by the baseline and context challenger.

Parameter	**Value**	**Purpose**
Objective / variance power	Tweedie / 1.3	Non-negative, right-skewed demand target
Boosting iterations	100	Fixed candidate complexity
Learning rate	0.035	Step size for boosting
Leaves / minimum child samples	31 / 80	Tree complexity and leaf support
Row / column subsampling	0.85 / 0.85	Stochastic regularisation
Subsample frequency	1	Apply row subsampling each iteration
L2 regularisation	2.0	Penalise excessive leaf weights
Random seed	42	Reproducibility of stochastic steps

The context challenger changes the information set, not the estimator family or hyperparameter budget. This controlled design makes the feature-group comparison easier to interpret: an observed difference is associated with the added historical context rather than a simultaneous change to the algorithm and its tuning configuration.

4.7 Historical Transaction-Feature Ablation

A separate seven-day ablation tests whether transaction-derived covariates reduce error beyond the historical demand features. Sales lines are joined to parent orders and the product master, then aggregated to shop–category–day for gross value, discount, order count, customer-days and quantity-weighted unit price. The aggregated quantity reproduces the complete observed shop–category panel exactly. The extended feature set uses only historical price, discount, order and customer-activity summaries from the 7-, 28- and 56-day windows before each forecast origin. It does not use future prices, promotion schedules, inventory outcomes or customer responses.

The ablation compares baseline LightGBM Tweedie with the same model plus these transaction-derived features at nine monthly origins from January through September 2025. Each origin covers all 512 series. Training windows use a 28-day stride and a minimum

56-day history; every training target window ends before its validation origin. The script and aggregate outputs are stored in research/forecasting/ablation_transaction_features.py and research/forecasting/results/transaction_feature_ablation_v1/. The ablation is development evidence, not an untouched final test.

4.8 Software Environment and Reproducibility

The experiment was executed with Python 3.13.5, pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0, and LightGBM 4.6.0. The source panel's SHA-256 checksum is recorded in the accompanying experiment manifest. The implementation writes the per-origin metrics, aggregate metrics, forecast-level predictions, demand-regime assignments, and regime-stratified metrics as machine-readable CSV files. The raw business dataset is retained in its controlled data location and is not copied into the public source repository.

4.9 Current Implementation Boundaries

The executable experiment remains a shop–category observed-sales forecast. It does not establish SKU-level latent demand, price or promotion uplift, or inventory-policy benefit. CatBoost Tweedie and a raw two-stage hurdle have now been evaluated as challengers; the hurdle probabilities remain uncalibrated. The inventory ledger has been reconciled internally, but negative-balance events and positive-sale days with non-positive intraday stock must be quarantined before SKU exposure labels are used. Calibrated prediction intervals, hierarchical reconciliation, point-in-time price and promotion features, and inventory replay remain subsequent tasks. The data are real business records according to the supplied manifest; however, real-data origin does not by itself establish generalisation. The current results are development estimates from selected periods. A final untouched period or held-out businesses are still required before claiming future predictive performance outside the development windows.

Chapter 5

Experimental Results

5.1 Evidence Boundary and Panel Integrity

The supplied dataset manifest identifies the package as real SME business data. The shop–category panel contains 561,152 observations, 512 unique series and 1,096 daily observations per series. It totals 1,580,361 units; 37.5962% of rows are explicit zero-sales observations. The audit found no duplicate shop–category–date keys, missing required values, zero-flag mismatches or missing dates within the complete panel. Transaction and inventory reconciliation findings are reported in Chapter 4.

5.2 Demand-Regime Distribution

Regimes are calculated using data available through 30 June 2025, before the first development origin. The ADI and CV^2 thresholds are 1.32 and 0.49. The panel contains 342 lumpy series, 153 erratic series, 13 smooth series and only 4 intermittent series.

Table 5.1: Historical demand-regime distribution across 512 shop–category series.

Regime	**Series**	**Share**	**Mean zero-sales ratio**
Lumpy	342	66.80%	47.62%
Erratic	153	29.88%	15.65%
Smooth	13	2.54%	3.64%
Intermittent	4	0.78%	83.94%
Total	512	100.00%	37.60%

Four intermittent series are not enough independent support for regime-specific automatic model switching. Repeated forecasts of the same four series do not create additional independent businesses.

5.3 Multi-Model Forecasting Performance

The fixed-origin experiment uses 1 July, 1 August and 1 September 2025. Each of the 512 series is forecast at each origin, giving 1,536 series-origin observations per horizon. The six candidates are LightGBM Tweedie, CatBoost Tweedie, a raw two-stage CatBoost hurdle, weekly seasonal-naive, Croston–SBA and TSB. All models use the same target definition and historical information at each origin.

TABLE 5.2: Pooled development WAPE by forecast horizon and candidate model.

Horizon	LightGBM WAPE	CatBoost WAPE	Hurdle WAPE	Croston–SBA WAPE	TSB WAPE	Seasonal naive
1 day	79.80%	79.98%	**79.23%**	84.01%	84.78%	92.21%
7 days	**33.74%**	34.20%	35.26%	37.74%	38.62%	40.77%
14 days	**24.11%**	24.78%	24.54%	30.27%	32.10%	33.97%
28 days	**18.57%**	19.28%	18.84%	25.62%	27.38%	29.16%

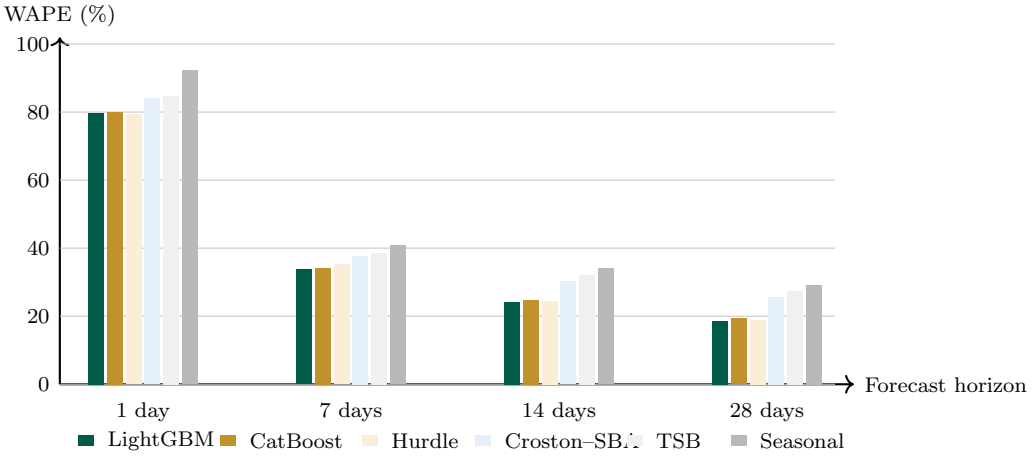


FIGURE 5.1: Measured development WAPE for the six candidates at four direct cumulative horizons. The one-day task remains difficult; lower WAPE at longer horizons reflects a different cumulative target and must not be interpreted as equivalent daily accuracy.

LightGBM has the lowest pooled WAPE at seven, fourteen and twenty-eight days. The raw hurdle candidate has the lowest one-day WAPE, but its one-day aggregate bias is −28.18%, compared with −21.69% for LightGBM. The hurdle probabilities are uncalibrated. Its lower one-day WAPE therefore does not justify promotion without calibration and a separate cost-sensitive analysis of under-forecasting.

Table 5.3: Forecast bias and horizon-scaled weekly MASE for the principal candidates.

Horizon	LightGBM bias	CatBoost bias	Hurdle bias	LightGBM MASE
1 day	−21.69%	−21.24%	−28.18%	0.793
7 days	−6.59%	−7.64%	−12.80%	0.326
14 days	−2.42%	−2.86%	−2.86%	0.226
28 days	−2.10%	−2.84%	−1.01%	0.175

Horizon-scaled weekly MASE divides absolute forecast error by the series-specific mean absolute lag-7 difference multiplied by the forecast horizon. It is a scale-normalised diagnostic, not a direct business-cost measure. The one-day WAPE remains high; longer-horizon results must not be interpreted as evidence of equally accurate daily replenishment forecasts.

5.4 Seven-Day Stability and Shop-Level Heterogeneity

LightGBM has the lowest seven-day pooled WAPE at each development origin: 32.52% on 1 July, 33.86% on 1 August and 34.92% on 1 September. CatBoost records 32.71%, 34.26% and 35.72%; the raw hurdle records 33.81%, 35.54% and 36.52%.

Table 5.4: Seven-day WAPE by development origin.

Origin	LightGBM	CatBoost	Hurdle	Croston–SBA	TSB
1 July 2025	**32.52%**	32.71%	33.81%	37.91%	39.82%
1 August 2025	**33.86%**	34.26%	35.54%	36.14%	35.82%
1 September 2025	**34.92%**	35.72%	36.52%	39.21%	40.24%

Across shops, LightGBM has lower seven-day WAPE than CatBoost in 22 of 32 shops; across categories it wins in 14 of 16. Median shop-level WAPE is 35.89% for LightGBM and 36.79% for CatBoost. The model is not uniformly best for every shop, so per-shop monitoring and abstention remain necessary.

Table 5.5: Cluster-bootstrap seven-day WAPE differences.

Challenger	**Median difference**	**95% interval**	**Interpretation**
CatBoost Tweedie	−0.45 pp	[−0.85, −0.08] pp	LightGBM lower WAPE
Raw CatBoost hurdle	−1.51 pp	[−2.12, −0.93] pp	LightGBM lower WAPE
Weekly seasonal-naive	−7.01 pp	[−8.55, −5.59] pp	LightGBM lower WAPE
Croston–SBA	−3.97 pp	[−5.04, −2.94] pp	LightGBM lower WAPE
TSB	−4.90 pp	[−6.11, −3.72] pp	LightGBM lower WAPE

The intervals resample 512 shop–category series with 2,000 bootstrap replicates. They quantify variation across the development series under the fixed-origin protocol; they are not uncertainty intervals for future real-world SME performance and do not replace an untouched test.

5.5 Historical Transaction-Feature Ablation

The transaction-derived feature experiment evaluates nine monthly development origins from January to September 2025, with 512 series at each origin and 4,608 series-origin forecasts per model. The baseline uses the historical demand features only. The extended candidate adds historical price, discount, order and customer-day summaries, all computed strictly before the forecast origin.

Table 5.6: Nine-origin seven-day LightGBM transaction-feature ablation.

Feature set	**Pooled WAPE**	**Mean origin WAPE**	**Bias**	**MAE**	**RMSE**
Historical demand only	**33.7313%**	**34.1557%**	-1.35%	**7.124**	**9.824**
Historical price/activity	33.8358%	34.2685%	-1.28%	7.146	9.866

The extended features improve WAPE at five origins and worsen it at four, but the pooled WAPE is 0.1045 percentage points higher and mean origin MAE is also higher. Aggregate under-forecast is marginally smaller (-1.28% versus -1.35%), showing that bias and absolute error do not always move together. The evidence does not support adding these transaction features to the current selected model. They are retained as a reproducible ablation, not adopted as a performance improvement. Future covariates should be added only when point-in-time availability is established and they improve pooled and origin-stability metrics.

5.6 Performance by Demand Regime

At seven days, CatBoost has slightly lower WAPE than LightGBM in the erratic group (28.50% versus 28.67%), while LightGBM performs better in the lumpy and smooth groups. Croston–SBA has lower WAPE in the intermittent group (77.02% versus 106.30% for LightGBM), but that subset contains only four series, twelve series-origin forecasts and 25 actual units. The raw hurdle has WAPE 83.18% in the same subset but under-forecasts total demand by 74.62%. None of these sparse subgroup results justifies automatic model routing.

Table 5.7: Seven-day WAPE by historical demand regime.

Regime	Cases	LightGBM	CatBoost	Hurdle	Croston–SBA
Erratic	459	28.67%	**28.50%**	28.57%	32.38%
Lumpy	1,026	**42.06%**	42.85%	45.60%	47.09%
Smooth	39	**19.89%**	22.03%	20.60%	20.69%
Intermittent	12	106.30%	121.62%	83.18%	**77.02%**

Table 5.8: Seven-day WAPE for the context challenger and baseline at each of nine monthly origins.

Origin	Baseline WAPE	Context WAPE	Change (pp)	Lower error
1 January 2025	34.34%	33.80%	-0.54	Context
1 February 2025	36.06%	35.54%	-0.52	Context
1 March 2025	31.07%	32.88%	+1.81	Baseline
1 April 2025	29.23%	28.36%	-0.87	Context
1 May 2025	31.90%	31.50%	-0.40	Context
1 June 2025	32.34%	33.50%	+1.17	Baseline
1 July 2025	40.83%	32.34%	-8.48	Context
1 August 2025	34.01%	33.71%	-0.29	Context
1 September 2025	34.82%	34.92%	+0.09	Baseline

The context features do not win at every origin. The largest improvement occurs in July, while March, June and September favour the baseline. This variation is important: a pooled reduction must not be presented as uniform improvement across time. It supports retaining the baseline as a fallback and monitoring the context model by origin and business.

5.7 Context-Aware Model Results

Table 5.9 compares the historical-feature baseline with the context-aware challenger on identical development origins. At seven days, pooled WAPE decreases from 33.47% to 32.66%, an absolute reduction of 0.81 percentage points (2.41% relative). The challenger wins at six of nine monthly origins. A paired 2,000-replicate shop–category cluster bootstrap

estimates the context-minus-baseline WAPE difference at -0.81 percentage points, with a 95% interval of [-1.09, -0.52] percentage points.

Table 5.9: Context-aware LightGBM versus the same-model historical-feature baseline.

Horizon	Baseline WAPE	Context WAPE	Absolute change	Context bias
1 day	82.73%	79.74%	-2.99 pp	-16.26%
7 days	33.47%	32.66%	-0.81 pp	-6.05%
14 days	28.48%	24.93%	-3.56 pp	+5.82%
28 days	19.35%	18.46%	-0.89 pp	+2.67%

The seven-day result is the strongest comparison because it uses nine origins; other horizons use three origins each. The context model's seven-day bias is -6.05%, compared with -2.02% for the baseline. A past-origin-only multiplicative adjustment reduced bias to -1.61% but increased WAPE to 32.90%. It is an optional cost-sensitive trade-off, not silently applied to the selected forecast.

Prequential scaled-residual intervals, calibrated only from earlier validation origins, achieved mean empirical coverage of 79.43% for nominal 80% intervals and 89.55% for nominal 90% intervals across the evaluated origins. These are development diagnostics, not guarantees of future coverage.

Bottom-up aggregation of the same seven-day leaf forecasts is coherent. WAPE is 32.66% at shop–category level, 14.95% at shop total, 10.98% at category total and 7.85% at network total. Lower aggregate errors partly reflect cancellation and do not establish equivalent leaf-level replenishment accuracy. The context model is the leading development challenger; production selection requires a genuinely untouched future period or held-out businesses and explicit under-forecast cost evaluation.

Table 5.10: Seven-day pooled error and bias diagnostics for the paired context experiment.

Metric	Baseline	Context model
Pooled WAPE	33.47%	32.66%
Mean origin WAPE	33.84%	32.95%
MAE (units per series-origin)	7.069	6.898
RMSE (units per series-origin)	10.008	9.792
Aggregate signed bias	-2.02%	-6.05%

The paired comparison uses the same 512 shop–category series at all nine origins. The 2,000-replicate cluster bootstrap resamples complete series clusters, retaining each sampled series' repeated-origin structure. The median WAPE difference (context minus baseline) is -0.811 percentage points, with a 95% interval from -1.091 to -0.517 percentage points. This interval quantifies variation across the development series; it is not a prediction interval for a future business or a substitute for the untouched final test.

Table 5.11: Prequential interval coverage for the seven-day context forecast.

Origin	Calibration origins	80% coverage	90% coverage	Interpretation
1 April 2025	3	75.78%	89.06%	80% under-covered
1 May 2025	4	84.77%	92.77%	Both above nominal
1 June 2025	5	71.29%	81.25%	Both under-covered
1 July 2025	6	82.81%	92.58%	Both above nominal
1 August 2025	7	82.62%	91.99%	Both above nominal
1 September 2025	8	79.30%	89.65%	Close to nominal

The interval method uses scaled absolute residuals from earlier origins only. Mean coverage across these six evaluated origins is 79.43% for the nominal 80% interval and 89.55% for the nominal 90% interval. Coverage nevertheless varies materially by month. The interval estimates are therefore useful diagnostics for uncertainty communication but are not yet sufficient to guarantee a fixed coverage level in deployment.

Table 5.12: Bottom-up seven-day forecast accuracy at different aggregation levels.

Aggregation level	Forecast cases	WAPE	Signed bias
Shop–category	4,608	32.66%	-6.05%
Shop total	288	14.95%	-6.05%
Category total	144	10.98%	-6.05%
Network total	9	7.85%	-6.05%

The hierarchy is coherent because shop, category and network totals are sums of the same shop–category leaf forecasts rather than separately fitted aggregate models. Error decreases as values are aggregated partly because positive and negative series errors can offset one another. Consequently, network-level accuracy cannot replace shop–category accuracy when the decision is a local reorder.

5.8 Model and Solution Design Decision

The current forecasting design retains the original pooled LightGBM Tweedie as a benchmark and advances the context-aware LightGBM as the leading development challenger. It uses the same model family and settings while adding historical shop, category and network aggregates. The nine-origin seven-day comparison shows lower pooled WAPE and a paired cluster-bootstrap interval favouring the context features. However, its greater under-forecast bias means the final decision must consider the business cost of shortages, not WAPE alone.

The current selection is therefore conditional: use the context-aware candidate for further validation, retain the baseline as fallback, and do not promote a production model until an untouched future period or held-out businesses have been evaluated. CatBoost remains a challenger; the hurdle model remains experimental until occurrence probabilities are calibrated and bias is controlled. Croston–SBA and TSB remain intermittent-demand comparators, but sparse regime support does not justify automatic routing.

SKU-level forecasting remains availability-aware. Missing SKU dates become zero only when active assortment and positive stock exposure are established. Negative-ledger events, stock-out periods, unknown assortment and closed-shop days remain censored or unknown. The separately identified simulated unmet-demand table is not used as observed sales or empirical lost-sales ground truth. RQ3 inventory-policy replay and RQ4 owner-facing explanation evaluation require their own evidence and experiments.

5.9 Threats to Validity and Summary

The results are measured on the supplied real-business dataset and are bounded to the shop–category panel, selected development origins and cumulative forecast horizons. The final untouched test has not yet been evaluated. Previously inspected Q4 2025 is excluded from model selection. The earlier WAPE values of 83.98%, 84.04% and 84.56% are not compared with the current experiments because their target, aggregation, origins and metric denominators have not been reproduced on identical evaluation rows.

The evidence supports advancing context features as a development challenger, while retaining the pooled baseline as fallback. Remaining work includes an untouched test, cost-sensitive bias evaluation, SKU exposure correction, interval validation and separate inventory-policy and owner-task experiments. No universal model winner or untested business-impact claim is asserted.

Chapter 6

Conclusions

6.1 Research Summary

This thesis develops and evaluates an integrated SME analytics framework using the supplied Bangladesh business records. The dataset manifest identifies the package as real SME sales and real people. Source tables were internally reconciled: sales lines join to orders, prepared SKU aggregates reproduce from transaction lines, category totals reproduce from SKU sales, and inventory sale movements match sales lines. The inventory audit also identifies negative-balance events and positive-sale days with non-positive reconstructed stock; these remain quarantined from SKU exposure labels pending correction.

For RQ1, the complete shop–category panel supports observed-sales forecasting. A context-aware LightGBM challenger adds historical shop, category and network aggregates to the baseline features. Across nine seven-day development origins, pooled WAPE improves from 33.47% to 32.66%, with a paired cluster-bootstrap interval supporting lower development error. However, under-forecast bias increases from -2.02% to -6.05%. The model is therefore a leading development challenger, not yet a final production selection. The one-, 14- and 28-day comparisons also show lower pooled WAPE, but use only three origins each.

Prequential interval diagnostics and bottom-up aggregation provide additional evidence. SKU latent demand, inventory-policy benefit, customer-action effectiveness and cross-sector transfer remain separate evaluation tasks. The contribution is an auditable, evidence-aware process that selects methods against common baselines, preserves temporal validity and links predictions to feasible business decisions. No accuracy target is forced and no untested business-impact claim is made.

Appendix A

RQ1 Data Audit and Target Eligibility

A.1 Analytical Grain and Field Definitions

The RQ1 source panel is defined at one row per shop, product category and calendar date. Its fields have the following roles:

Table A.1: Field dictionary for the complete RQ1 panel.

Field	Type	Meaning and validation
`shop_id`	Categorical key	Shop identifier; retained as a model feature and grouping key
`category`	Categorical key	Product category; retained as a model feature and grouping key
`date`	Date	Calendar date; parsed and checked for daily continuity
`units_sold_observed`	Non-negative numeric	Observed units sold in the shop-category-date cell; target quantity
`is_zero_observed_sales`	Binary indicator	Must equal one exactly when observed units are zero

The complete panel contains 512 shop–category series. Each series has 1,096 consecutive dates from 1 January 2023 to 31 December 2025. The resulting row count is therefore $512 \times 1{,}096 = 561{,}152$. The total observed quantity is 1,580,361 units. Explicit zero-sales rows number 210,972, giving a zero-sales share of 37.5962%.

A.2 Observed Zero, Missing Observation and Censored Demand

Three states must be kept separate:

1. **Observed zero:** a complete shop–category–date row exists, its quantity is zero and the zero indicator agrees. This is a valid observed-sales target for the current RQ1 task.

2. **Missing observation:** the expected row is absent. Its meaning is unknown until source completeness, operating calendar and assortment are checked. It is not automatically zero.

3. **Censored demand:** a product may have been wanted by customers but unavailable because of stock-out or other operating constraints. Recorded sales may then understate latent demand.

The complete shop–category panel passes the daily continuity and zero-indicator checks. The sparse SKU export does not provide a complete SKU-date panel. A missing SKU date can therefore represent inactive assortment, a closed shop, an incomplete export or an unavailable product. SKU-level latent-demand modelling is not enabled until product activity, shop operation and stock exposure are validated for each date.

The inventory reconstruction identified 269 negative-balance movement events across 253 shop–SKU combinations and 330 positive-sale days with non-positive reconstructed intraday stock. These events are retained as exceptions and excluded from any claim that the affected SKU-day was continuously in stock. The 2,088 theoretical shop–SKU combinations without an opening movement are not assumed to be active series.

A.3 Audit Rules and Exception Handling

The ingestion checks are deterministic and are run before model training. A failed check blocks the affected analytical task or quarantines the affected records. The procedure is summarised below.

Table A.2: RQ1 data-audit rules and failure response.

Rule	**Pass condition**	**Failure response**
Schema	All required fields are present with expected types	Stop ingestion and report missing or malformed fields
Key uniqueness	One row per shop-category-date	Reject duplicate keys; do not silently aggregate
Quantity validity	Quantity is finite and non-negative	Quarantine invalid rows and reconcile with source transactions
Zero indicator	Indicator equals the zero test on quantity	Reject the panel until the discrepancy is resolved
Calendar completeness	Every series contains the same daily calendar	Investigate missing dates; do not fill with assumed zero
Order linkage	Each sales line has a valid parent order and matching shop	Quarantine broken references
Inventory linkage	Sale movement matches order, shop, SKU, lot and quantity	Quarantine affected stock-exposure labels

The audit is deliberately layered. Panel integrity establishes whether the forecast target is structurally usable. Transaction reconciliation tests whether the prepared daily aggregates reproduce from source sales lines. Inventory reconciliation assesses whether the ledger can support availability-aware SKU targets. Passing one layer does not imply that all other layers have passed.

Appendix B

RQ1 Feature Engineering and Model Specification

B.1 Historical Feature Definitions

For a series with observed daily sales y_t, all features are calculated using observations strictly before the forecast origin t. The principal feature definitions are:

$$\mathrm{Lag}_k(t) = y_{t-k}, \qquad k \in \{1, 7, 14, 28, 56\}.$$

For a rolling window of length w,

$$\mathrm{Mean}_w(t) = \frac{1}{w}\sum_{j=1}^{w} y_{t-j}, \qquad \mathrm{Sum}_w(t) = \sum_{j=1}^{w} y_{t-j}.$$

The rolling standard deviation is calculated over the same historical window. The zero-sales ratio is

$$\mathrm{ZeroRatio}_w(t) = \frac{1}{w}\sum_{j=1}^{w} \mathbb{1}(y_{t-j} = 0).$$

Let N_t^+ denote the number of positive-sales days before origin t, and let T_t denote the length of the observed history. The average demand interval is

$$\mathrm{ADI}_t = \frac{T_t}{N_t^+},$$

when at least one positive-sales day exists. Positive-demand mean and variance are calculated only over positive observations. The squared coefficient of variation is

$$CV_t^2 = \frac{\mathrm{Var}(y_t \mid y_t > 0)}{\mathrm{E}(y_t \mid y_t > 0)^2},$$

when the positive-demand mean is non-zero. Days since last sale is the number of complete days between the most recent positive observation and the forecast origin.

B.2 Context Feature Construction

The context challenger adds historical aggregates without changing the forecast target. For shop b, category c, and window w, the shop aggregate is

$$S_{b,w}(t) = \sum_{j=1}^{w} \sum_{c'} y_{b,c',t-j},$$

and the category aggregate is

$$C_{c,w}(t) = \sum_{j=1}^{w} \sum_{b'} y_{b',c,t-j}.$$

The network aggregate is

$$N_w(t) = \sum_{j=1}^{w} \sum_{b'} \sum_{c'} y_{b',c',t-j}.$$

The model also receives recent shop/category trend summaries and the focal series' share of its shop and category totals. These features describe relative scale and recent movement across related series. All aggregate windows terminate at $t-1$; no observation from the forecast target window is included.

The context experiment is an ablation of information availability: the baseline receives own-series history and calendar features, while the challenger receives those same features plus cross-series context. The estimator family, hyperparameters, target, origins and evaluation rows are held constant.

B.3 Direct Multi-Horizon Target

For each horizon $h \in \{1, 7, 14, 28\}$, the target is the sum of observed units in the next h days:

$$Y_{s,t,h} = \sum_{j=0}^{h-1} y_{s,t+j}.$$

A direct model is fitted for each horizon. This means the one-day target and 28-day target are different prediction tasks. A lower WAPE at a longer cumulative horizon must not be interpreted as equivalent accuracy for daily replenishment.

Historical training origins are sampled on a fixed stride. A training sample is admitted only when its full target window ends before the validation origin. The minimum historical feature window is 56 days. This chronological condition is applied to both the baseline and context-aware models.

B.4 Model Configuration and Reproducibility

The context experiment uses LightGBM Tweedie with variance power 1.3, 100 boosting iterations, learning rate 0.035, 31 leaves, minimum child samples 80, row and column subsampling of 0.85, subsampling frequency 1, L2 regularisation 2.0 and random seed 42. The configuration is fixed between baseline and context model.

The seven-day experiment uses nine monthly origins from January through September 2025. The one-, 14- and 28-day comparisons use July, August and September 2025. Each origin contains all 512 shop–category series. The previously inspected Q4 2025 window is excluded from model selection. The run manifest records the input checksum, evidence status, origins, horizons, feature groups, parameters, bootstrap settings and the fact that no untouched final test has been evaluated.

Appendix C

RQ1 Evaluation and Model Selection Record

C.1 Metric Definitions

For observed values y_i and predictions $\hat{y}_i$, the principal metrics are:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|,$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

$$\text{WAPE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{\sum_{i=1}^{n}|y_i|},$$

and signed aggregate bias is

$$\text{Bias} = \frac{\sum_i \hat{y}_i - \sum_i y_i}{\sum_i y_i}.$$

WAPE summarises absolute error relative to total observed volume. MAE expresses average error in the target's units, while RMSE gives greater weight to large errors. Bias identifies systematic over-forecasting or under-forecasting. No single metric is sufficient for model selection because a candidate may reduce absolute error while increasing under-forecasting.

C.2 Paired Cluster Bootstrap

The seven-day context comparison contains repeated forecasts for the same 512 shop–category series across nine origins. Resampling individual series-origin rows would ignore within-series dependence. Instead, the bootstrap samples 512 complete shop–category clusters with replacement. For each replicate, all origins belonging to a sampled series are retained. The WAPE difference is calculated as

$$\Delta_{\mathrm{WAPE}} = \mathrm{WAPE}_{\mathrm{context}} - \mathrm{WAPE}_{\mathrm{baseline}}.$$

A negative value favours the context model. The analysis uses 2,000 replicates and a fixed seed. The median difference is -0.811 percentage points and the 95% percentile interval is [-1.091, -0.517] percentage points. This interval describes development-series variation and does not replace evaluation on a future period or independent businesses.

C.3 Prequential Uncertainty Assessment

Prediction intervals are assessed prequentially: residuals from earlier validation origins calibrate the interval for a later origin. Residuals from the current or future origin are not used to calibrate its own interval. Residuals are scaled by the square root of the forecast level with a lower bound to avoid division by zero. Empirical coverage is reported separately by origin because average coverage can conceal period-specific under-coverage.

Across the evaluated April–September origins, mean empirical coverage is 79.43% for nominal 80% intervals and 89.55% for nominal 90% intervals. The variation between origins means that these intervals remain development diagnostics. Before operational use, the calibration procedure must be evaluated on an untouched period and monitored for coverage drift.

C.4 Hierarchical Coherence

The reported shop, category and network totals are formed by summing the same shop–category forecasts. This bottom-up construction guarantees arithmetic coherence: the shop and category totals reconcile to the network total. It does not guarantee that the forecast is unbiased or optimal at every level. Aggregation can reduce percentage error because over- and under-forecasts across series may cancel.

The seven-day WAPE decreases from 32.66% at shop–category level to 14.95% at shop total, 10.98% at category total and 7.85% at network total. Local stock decisions must still use shop-level or category-level evidence appropriate to the decision; network accuracy cannot be substituted for a local reorder forecast.

C.5 Model Selection Decision and Remaining Gates

The current evidence supports advancing the context-aware LightGBM as the leading development challenger while retaining the historical-feature LightGBM as fallback. The context model improves pooled WAPE at the four tested horizons, and the seven-day comparison is supported by nine origins and a paired cluster bootstrap. However, seven-day under-forecast bias increases from -2.02% to -6.05%. A past-origin-only multiplicative adjustment reduces bias but increases WAPE. The final choice therefore depends on the business cost of shortages and cannot be made from WAPE alone.

The following gates remain before production selection:

1. Freeze the feature definitions, training procedure and primary metrics.
2. Evaluate on a genuinely untouched future period or held-out businesses.
3. Compare forecast error and bias under business-relevant shortage and holding-cost assumptions.
4. Verify prediction-interval coverage on the untouched period.
5. Complete SKU assortment and stock-exposure validation before SKU-level latent-demand claims.
6. Evaluate inventory-policy outcomes separately from forecast accuracy.
7. Assess owner-facing recommendations through separate decision tasks.

Until these gates are passed, the model is a validated development challenger, not a claim of universal superiority or proven inventory savings.

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